

A Secular Mass-Ratio Construction for the Electroweak Pole Spectrum: String, Kaluza–Klein, and G_2 Interpretation Targets

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The Rivero–de Vries electroweak observation is formulated as a pole-spectrum mass-ratio problem organized by a two-branch secular equation. The algebraic input inserts $J = s(s+1)$, historically a quadratic-Casimir eigenvalue, into $x^2 + Jx - J = 0$. The positive-root quotient $1 - x_+(3/4)/x_+(2) = 0.22310132\dots$ is proposed for comparison with the W/Z complex-pole mass-ratio observable s_{pole}^2 , with the low-energy pole spectrum taken as the working placement hypothesis. The negative branch belongs to the same diagonalization and is treated as a candidate scalar/order-parameter datum whose physical meaning requires a gauge-invariant scalar functional. Endpoint, interval Kaluza–Klein, localized G_2 , and electroweak-superconnection settings supply candidate arenas for a two-channel kernel. The unresolved derivation is the ordered assignment $(J_W, J_Z) = (3/4, 2)$, the source of the DeVries kernel entries, and the pole-matched scalar interpretation of the partner branch.

I. INTRODUCTION

The electroweak sector contains a dimensionless pole-spectrum ratio,

$$s_{\text{pole}}^2 = 1 - \frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2}, \quad (1)$$

which is fixed experimentally by the pole positions of the charged and neutral weak vector bosons. In the Standard Model at tree level the same ratio is written as an angular datum in coupling space, while the Higgs vacuum expectation value fixes the common radial scale of the vector masses. This separation between projective and radial data gives the working language of the present paper.

The Rivero–de Vries observation inserts $J = s(s+1)$, a quadratic-Casimir eigenvalue, into a dimensionless two-channel ansatz. In the normalized notation used below, the equation is

$$x^2 + Jx - J = 0, \quad J = s(s+1). \quad (2)$$

The positive branch gives

$$s_{\text{dV}}^2 = 1 - \frac{x_+(3/4)}{x_+(2)} = 0.22310132\dots \quad (3)$$

The original note emphasized the proximity of this number to the mass-shell weak angle and motivated the operator through Casimir invariants and high-spin behavior [1]. In this paper Eq. (3) is used as a descriptive spectral clue. The source-theory derivation of the two-channel ansatz is open. The central question asks which physical object carries the square root in Eq. (2), and why that object is read at the low-energy W/Z pole ratio.

This placement question is the conceptual core. A standard grand-unified lineage uses representation theory to obtain a clean weak-angle value at a high scale. In SU(5), and in Spin(10) with the usual fermion content, the famous group-theoretic value is $\sin^2 \theta_W = 3/8$; contact with the observed low-energy angle is then made

through running and thresholds [24]. The Rivero–de Vries construction places the scale burden on the pole-mass ratio itself. The required source theory must explain why a spectral datum attaches to the pole spectrum. A unification-scale boundary condition would define a different reading of the same algebra.

The string-theoretic setting uses the older dual-model and Regge ambition as part of the physical context. Open strings, endpoint data, Kaluza–Klein spectra, and boundary conditions are treated as possible mechanisms for organizing low-energy particle structure. String theory also incorporates Kaluza–Klein ideas, grand-unified representations, and supersymmetric structure in transmuted form [3]. The physical question is whether those tools can produce an electroweak secular datum at the observed pole spectrum. The numerical clue remains a clue until that mechanism is written.

The construction has four layers. The first layer is algebraic: a two-channel secular equation has roots whose ratio is fixed by the Casimir variable. The second layer is electroweak: the W/Z mass ratio is a projective datum along the broken-to-unbroken ray of the gauge-Higgs system. The third layer is spectral: the square root must be assigned to a pole, running, compactification, or boundary-condition object. The fourth layer is dynamical: strings, branes, Kaluza–Klein boundary conditions, or singular G_2 geometries may generate the secular equation or its branch structure.

The logical structure used below separates three assumptions from four theorem targets. Assumption A1 fixes $Q(J)$ as the algebraic two-channel object. Assumption A2 reads s_{pole}^2 with complex pole masses. Assumption A3 sets the ordered electroweak assignment $(J_W, J_Z) = (3/4, 2)$. Target I places the quotient in the pole scheme; Target II derives A3 from gauge-Higgs or source data; Target III derives the route-kernel entries; Target IV identifies the x_- branch. The algebraic calculation is explicit. The pole-mass comparison is a placement hypothesis whose scheme conventions are fixed in Sec. III. In this paper the ordered assignment

A3 is an explicit working assumption for the conditional pole-ratio proposal. Its derivation is retained as Target II: a source package must derive the charged order-parameter sample $J_H = 3/4$, the photon-orthogonal neutral current sample $J_{\text{adj}} = 2$, and the pole-scheme remainder before the relation becomes a source-derived electroweak result. The negative branch is kept as a candidate scalar/order-parameter datum. The electromagnetic coupling α lies outside the claim of this paper. A claim for α would require a source-fixed scalar normalization, a derived $U(1)_{\text{em}}$ generator and charge lattice, and a specified pole/running matching scale.

Target II has an electroweak-superconnection formulation. In that formulation the odd Higgs/order-parameter component is projected to $J_H = 3/4$, while the even transverse current component is projected to $J_{\text{adj}} = 2$:

$$P_W : \Phi_{\text{odd}} \mapsto J_H = \frac{3}{4}, \quad P_Z : F_{\text{even}}^{\gamma^\perp} \mapsto J_{\text{adj}} = 2.$$

A successful version uses one gauge-Higgs complex, one inner product, and one pole-matching map. Coquereaux's $SU(2|1)$ superconnection source supplies the electroweak grading vocabulary [39]; the ordered pole assignment remains the theorem target.

The paper is organized as follows. Section II gives the secular operator and its roots. Section III defines the pole observable and the placement problem. Section IV formulates the projective electroweak interpretation. Section V records the negative branch and its possible Higgs-scale role. Sections VI–XIII develop the string, endpoint, Kaluza–Klein, and G_2 mechanism tests in a common two-channel language. Sections XIV and XV delimit the flavor and global-gauge-group issues. Section XVI gives the conclusion and outlook.

II. TWO-CHANNEL SECULAR MASS-RATIO OPERATOR

Let $J = s(s+1)$ be the historical quadratic-invariant input. The normalized two-channel operator used in this paper is

$$\frac{Q(J)}{\mu^2} = \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}. \quad (4)$$

Its characteristic equation is

$$\det \left(xI - \frac{Q(J)}{\mu^2} \right) = x^2 + Jx - J = 0. \quad (5)$$

The two roots are

$$x_{\pm}(J) = \frac{-J \pm \sqrt{J^2 + 4J}}{2}. \quad (6)$$

They obey

$$x_+(J) + x_-(J) = -J, \quad x_+(J)x_-(J) = -J. \quad (7)$$

For $J > 0$, the positive branch is positive and the negative branch is negative.

Historically, the electroweak number is obtained by sampling the positive branch at the two Casimir values

$$J_H = \frac{3}{4}, \quad J_{\text{adj}} = 2. \quad (8)$$

This gives

$$s_{\text{dV}}^2 = 1 - \frac{x_+(3/4)}{x_+(2)}. \quad (9)$$

The expression is dimensionless; the overall scale μ cancels. The cancellation is essential for an electroweak interpretation because Eq. (9) can then be compared with a projective W/Z mass ratio. In the electroweak reading developed here, $J_H = 3/4$ is the Higgs/order-parameter doublet invariant and $J_{\text{adj}} = 2$ is the adjoint/current invariant. The ordered use of these two samples in the W/Z quotient remains the central derivation problem.

The relation to the original Rivero operator is obtained by restoring the Poincare Casimir eigenvalues $C_1 = m^2$ and $C_2 = -m^2 s(s+1)$. In the original construction, a mass-squared operator is formed from combinations of Casimir invariants, with a high-spin condition designed to preserve the asymptotic mass behavior relevant for Regge trajectories [1]. Equation (4) is the reduced dimensionless representative of that branch problem. Its use here requires a later source-theory derivation; Casimir language identifies the input datum and the historical clue.

A. What the algebra fixes

The operator in Eq. (4) fixes three algebraic facts. First, the trace is $-J$. Second, the determinant is $-J$. Third, the two eigenvalues are tied by the same invariant:

$$\text{tr} \frac{Q(J)}{\mu^2} = -J, \quad \det \frac{Q(J)}{\mu^2} = -J. \quad (10)$$

Thus a physical derivation has to explain a simultaneous trace and determinant assignment. A model that gives only a square-root branch, or only a J -dependent diagonal term, leaves the DeVries construction underdetermined. The required two-channel matrix has the schematic form

$$\mathcal{M}_J = \begin{pmatrix} 0 & \kappa_J \\ \kappa_J & -\tau_J \end{pmatrix}, \quad \kappa_J^2 = J, \quad \tau_J = J. \quad (11)$$

This is the compact target for the endpoint, Kaluza–Klein, and G_2 sections.

Trace and determinant statements in this paper are matrix-specific. The trace of $Q(J)$, the trace of any first-seed kernel $K_{\text{seed}}^{(1)}$, and the trace of any σ_3 -projected negative-sector kernel $K_-^{\sigma_3}$ belong to distinct vector

spaces until a common source operator, field basis, projection map, and normalization identify them. A coefficient transfer between these matrices requires an explicit source-theory theorem.

The square-root radical is then the discriminant of this two-channel problem:

$$\Delta_J = J^2 + 4J. \quad (12)$$

The radical has to attach to a physical object. The radical-placement section lists the main options. This paper follows the pole-spectrum placement because the positive and negative branches are interpreted as a single low-energy spectral clue.

B. Minimal assumptions and theorem target

A source derivation must state the minimal assumptions that select Eq. (4). The current minimal list is:

1. dependence on available Poincare or internal invariants;
2. dimensional consistency after restoring the mass scale μ ;
3. a two-branch secular structure;
4. preservation of the high-spin or Regge asymptotic datum in the originating problem;
5. a trace/determinant condition equivalent to Eq. (10).

These assumptions define the uniqueness target. The theorem to prove would read: within a stated class of two-channel operators built from the allowed invariants, the high-spin condition and trace/determinant constraints select Eq. (4) up to similarity transformations and normalization.

This theorem would also clarify the status of the negative branch. Since $x_-(J)$ is fixed by the same trace and determinant as $x_+(J)$, a derivation of Eq. (4) automatically derives both branches. The electroweak interpretation can then ask which fields or boundary data realize the two eigenvectors.

III. POLE OBSERVABLE AND ELECTROWEAK SCHEME SEPARATION

The observable used for comparison is the pole-mass ratio

$$s_{\text{pole}}^2 = 1 - \frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2}. \quad (13)$$

This ratio is distinct from a running $\overline{\text{MS}}$ weak angle and from effective weak mixing angles extracted from

neutral-current asymmetries. This paper therefore treats Eq. (13) as the primary target and later relates other schemes to it.

The following placement convention will be used. The Rivero-de Vries radical is read as a spectral object attached to the W/Z pole data. Other readings could attach the same radical to a high-scale boundary value, to a running $\overline{\text{MS}}$ coupling, or to an internal compactification eigenvalue. Each reading changes the physical content of the proposal. The pole reading gives the word “spectrum” its literal electroweak meaning.

A vector-boson pole is defined by the complex pole of the propagator,

$$s_V = M_{V,\text{pole}}^2 - iM_{V,\text{pole}}\Gamma_{V,\text{pole}}. \quad (14)$$

Throughout this paper, $M_{V,\text{pole}}$ and $\Gamma_{V,\text{pole}}$ refer to the parameterization in Eq. (14). Martin and Robertson also record the alternative reporting convention

$$s_{\text{pole}} = (M - i\Gamma/2)^2. \quad (15)$$

Any input reported in that convention must be translated to Eq. (14) before Eq. (13) is formed [10]. For an unmixed charged vector, s_V is the zero of the inverse transverse propagator after renormalization,

$$\Delta_{T,V}^{-1}(s_V) = 0. \quad (16)$$

In the neutral sector, s_Z is the massive eigenvalue pole of the transverse (γ, Z) inverse-propagator matrix. Equivalently, after the photon null direction is separated,

$$\det \Delta_{T,\gamma Z}^{-1}(s_Z) = 0 \quad \text{on the massive neutral subspace.} \quad (17)$$

When experimental inputs are quoted in a variable-width Breit–Wigner line-shape convention, the conversion to pole parameters must be performed before using Eq. (13). With a common W/Z variable-width convention, the following equations define a conditional convention map with γ_V formed from the quoted variable-width parameters:

$$\gamma_V = \frac{\Gamma_{V,\text{BW}}}{M_{V,\text{BW}}}. \quad (18)$$

The pole parameters in Eq. (14) are then obtained as

$$M_{V,\text{pole}} = \frac{M_{V,\text{BW}}}{\sqrt{1 + \gamma_V^2}}, \quad \Gamma_{V,\text{pole}} = \frac{\Gamma_{V,\text{BW}}}{\sqrt{1 + \gamma_V^2}}. \quad (19)$$

Martin and Robertson summarize the corresponding vector-boson mass relation as

$$M_{V,\text{BW}}^2 = M_{V,\text{pole}}^2 + \Gamma_{V,\text{pole}}^2, \quad V = W, Z. \quad (20)$$

Equations (19)–(20) apply after the quoted W and Z inputs have been verified to share the same variable-width Breit–Wigner convention. Fixed-width line-shape inputs, already-pole inputs, and inputs using Eq. (15) require their stated convention map. This conversion is part of the source check for the pole comparison [10].

The resulting convention chain is

$$(M_{V,\text{BW}}, \Gamma_{V,\text{BW}}) \longrightarrow (M_{V,\text{pole}}, \Gamma_{V,\text{pole}}) \longrightarrow s_V \longrightarrow s_{\text{pole}}^2. \quad (21)$$

A. Scheme map

The comparison uses the following scheme map before any weak-angle quantity is interpreted:

Object	Definition level	Role in this paper
s_{pole}^2	complex-pole mass ratio	primary low-energy spectral target
Breit–Wigner masses	experimental line-shape convention	input convention requiring Eq. (20)
$\hat{s}^2(Q)_{\overline{\text{MS}}}$	running coupling ratio in a renormalization scheme	translation and loop-scale comparison
$\sin^2 \theta_{\text{eff}}^\ell$	neutral-current effective angle	phenomenological comparison after electroweak corrections
GUT boundary value	high-scale representation datum	comparison for structural weak-angle values
compactification eigenvalue	internal spectral datum	possible origin for $\mathcal{O}_J$ after a dynamical derivation

The pole reading selects the first row. The remaining rows are translation targets. The source-checked electroweak comparison begins from the quoted experimental convention, maps to complex pole quantities, forms Eq. (13), and then compares with Eq. (9).

The inherited numerical illustration uses

$$M_W = M_Z \sqrt{1 - s_{\text{dV}}^2}. \quad (22)$$

Appendix B records the inherited arithmetic for provenance.

B. PDG W/Z input set

The Particle Data Group live tables and 2025 listings quote the W and Z masses as mass-dependent-width Breit–Wigner parameters. The W listing gives $M_{W,\text{BW}} = 80.3692 \pm 0.0133 \text{ GeV}$ and $\Gamma_{W,\text{BW}} = 2.14 \pm 0.05 \text{ GeV}$. The Z listing gives $M_{Z,\text{BW}} = 91.1879 \pm 0.0020 \text{ GeV}$ and $\Gamma_{Z,\text{BW}} = 2.4955 \pm 0.0023 \text{ GeV}$ [12, 13]. Applying Eq. (19) gives the descriptive pole inputs

Boson	M_{BW} (GeV)	Γ_{BW} (GeV)	M_{pole} (GeV)	Γ_{pole} (GeV)
W	80.3692 ± 0.0133	2.14 ± 0.05	80.3407 ± 0.0134	2.1392 ± 0.0499
Z	91.1879 ± 0.0020	2.4955 ± 0.0023	91.1538 ± 0.0020	2.4946 ± 0.0023

The uncertainty propagation in the last two columns treats the quoted mass and width errors as uncorrelated. Correlations in a future electroweak-average combination would change the uncertainty assignment, while the convention map itself is fixed by Eq. (19).

With the same uncorrelated-input convention, the descriptive pole-ratio value is

$$(s_{\text{pole}}^2)_{\text{PDG input}} = 0.2231768 \pm 0.0002608. \quad (23)$$

This is the W/Z input line used for descriptive comparison in this paper. The CDF-II measurement and the CMS 2026 measurement $M_W = 80.3602 \pm 0.0099 \text{ GeV}$ are important W-mass inputs [14, 15]; they are not used as standalone replacements for Eq. (23) without a common Z input, width convention, covariance prescription, and averaging rule.

The Standard Model parameter literature distinguishes several vacuum and renormalization conventions. In particular, pure $\overline{\text{MS}}$ treatments may define the Higgs vacuum expectation value as the minimum of the loop-corrected effective potential in Landau gauge, eliminating tadpole diagrams at the cost of gauge-choice dependence in intermediate parameters [10]. Such scheme details are relevant for comparing a mass-ratio statement with Lagrangian parameters, while Eq. (13) remains the pole-spectrum target.

Loop effects are therefore a conceptual discriminator. Pole, effective, and running definitions are related by scheme-dependent self-energies and threshold terms. They ask what object the radical belongs to and at what scale it is to be read. A high-scale reading would make the DeVries value analogous to a GUT boundary datum. The pole-data reading makes it a low-energy spectral datum and transfers the burden to string/Kaluza–Klein dynamics.

C. Limited claim

The formal claim of this section is limited:

$$s_{\text{dV}}^2 \text{ is compared first with } 1 - \frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2}. \quad (24)$$

Every comparison to $\overline{\text{MS}}$ inputs, effective angles, or line-shape masses requires an explicit map. Thus a disagreement among schemes diagnoses the placement of $\sqrt{J^2 + 4J}$ and the scale at which the radical is read.

IV. ELECTROWEAK EMBEDDING AS A PROJECTIVE DATUM

The gauge-Higgs sector gives the tree-level vector masses from the Higgs kinetic term

$$\begin{aligned}
 & (D_\mu H)^\dagger (D^\mu H), \\
 D_\mu &= \partial_\mu - \frac{ig}{2} \sigma^a W_\mu^a - \frac{ig'}{2} B_\mu, \\
 H &= \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h \end{pmatrix}.
 \end{aligned} \quad (25)$$

The charged fields

$$W_\mu^\pm = \frac{1}{\sqrt{2}} (W_\mu^1 \mp iW_\mu^2) \quad (26)$$

receive a mass term $g^2 v^2 W_\mu^+ W^{-\mu}/4$. The neutral fields have mass matrix

$$\frac{v^2}{8} \begin{pmatrix} W_\mu^3 & B_\mu \end{pmatrix} \begin{pmatrix} g^2 & -gg' \\ -gg' & g'^2 \end{pmatrix} \begin{pmatrix} W^{3\mu} \\ B^\mu \end{pmatrix}. \quad (27)$$

It is diagonalized by

$$\begin{aligned} Z_\mu &= \cos \theta_W W_\mu^3 - \sin \theta_W B_\mu, \\ A_\mu &= \sin \theta_W W_\mu^3 + \cos \theta_W B_\mu, \\ \tan \theta_W &= \frac{g'}{g}. \end{aligned} \quad (28)$$

This gives

$$M_W^2 = \frac{g^2 v^2}{4}, \quad M_Z^2 = \frac{(g^2 + g'^2) v^2}{4}, \quad M_\gamma^2 = 0. \quad (29)$$

The ratio is

$$1 - \frac{M_W^2}{M_Z^2} = \frac{g'^2}{g^2 + g'^2}. \quad (30)$$

Thus the vacuum scale v fixes the radial size of the massive vector pair, while the quotient fixes a projective direction in the two-coupling plane [5, 9].

A. Custodial structure and the electroweak ray

The Standard Model doublet can be packaged as the bidoublet

$$\Phi_H = \begin{pmatrix} H^{0*} & H^+ \\ -H^{+*} & H^0 \end{pmatrix}. \quad (31)$$

In the scalar sector this makes the approximate global $SU(2)_L \times SU(2)_R$ structure visible; the vacuum proportional to the identity preserves the diagonal custodial $SU(2)$ subgroup [9]. Hypercharge gauging and Yukawa splittings break the symmetry, while the tree-level doublet mass matrix preserves the relation

$$\rho = \frac{M_W^2}{M_Z^2 \cos^2 \theta_W} = 1. \quad (32)$$

This is the electroweak reason the DeVries comparison is read as a projective statement about the W/Z pair. The broken-to-unbroken deformation scales the vector masses with v while preserving the coupling-space direction.

The DeVries proposal is read in the same projective language. In a source route the pole statement has the form

$$\frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2} = \frac{x_+(3/4)}{x_+(2)} + \Delta_{O1}. \quad (33)$$

Here Δ_{O1} collects field normalization, threshold, source-scale, and complex-pole matching data. The tree-level analogy is the pre-pole comparison

$$\Delta_{O1}^{\text{tree}} = \frac{g^2}{g^2 + g'^2} - \frac{x_+(3/4)}{x_+(2)}, \quad (34)$$

using Eq. (29). A source theorem must derive the value of Δ_{O1} in the complex-pole scheme. Sec. III and Appendix C 1 state that matching obligation. The common scale is then supplied by the electroweak vacuum or, in a deeper model, by the compactification or brane scale that generates the effective gauge-Higgs sector.

The unbroken limit is the ray

$$v \rightarrow 0 \quad \text{with} \quad \frac{g'^2}{g^2 + g'^2} \text{ fixed}. \quad (35)$$

Along this ray, M_W and M_Z vanish together and the photon remains massless. This is the electroweak deformation relevant to the construction. The projective datum remains meaningful as the limiting direction even when the radial mass scale is taken to zero.

Equivalently, introduce a parameter t by

$$\begin{aligned} H_t &= \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ tv + h \end{pmatrix}, \\ g(t) &= g, \quad g'(t) = g', \quad 0 \leq t \leq 1. \end{aligned} \quad (36)$$

Then

$$\begin{aligned} M_W^2(t) &= t^2 \frac{g^2 v^2}{4}, \\ M_Z^2(t) &= t^2 \frac{(g^2 + g'^2) v^2}{4}, \\ M_\gamma^2(t) &= 0. \end{aligned} \quad (37)$$

This is a tree-level ray statement before the pole-matching map of Sec. III. Coupling-space paths such as $g \rightarrow 0$, $g' \rightarrow 0$, $g'/g \rightarrow 0$, or $g'/g \rightarrow \infty$, and custodial-breaking shifts such as $\delta M_W^2 \neq \delta M_Z^2 \cos^2 \theta_W$, define separate deformation problems. A source route for the DeVries determinant must preserve the single radial parameter, the photon null direction, and the projective W/Z direction. Routes with different behavior enter as failure modes or as separate models.

The ordered assignment used in assumption A3 is

$$(J_W, J_Z) = \left(\frac{3}{4}, 2 \right). \quad (38)$$

The electroweak gauge fields are gauge-algebra fields before symmetry breaking: W^1, W^2, W^3 live in the adjoint of $SU(2)_L$, while B is the gauge field of the abelian $U(1)_Y$ factor. The value $J = 2$ has a direct $SU(2)_L$ adjoint reading through $T(T+1)$ with $T = 1$. The value $J = 3/4$ belongs to the Higgs/order-parameter side, where the symmetry-breaking field is an $SU(2)_L$ doublet with $T = 1/2$ and $T(T+1) = 3/4$.

These representation labels define the available samples:

$$J_Z = T_{\text{adj}}(T_{\text{adj}} + 1) = 2, \quad J_H = T_H(T_H + 1) = \frac{3}{4}. \quad (39)$$

The source-completion target is the ordered sampling rule that turns the pair (J_H, J_{adj}) into the W/Z quotient in Eq. (33). A derivation must start from the gauge-Higgs Lagrangian, a current-algebra construction, or a compactification boundary problem and produce Eq. (38) from the same determinant that supplies the negative branch.

B. The assignment problem as a theorem target

The theorem target can be stated mechanism-independently. Let $\mathcal{E}$ be the electroweak gauge-Higgs system, including its projective W/Z pole data and scalar order-parameter channel. A DeVries derivation is a map

$$\mathcal{D} : \mathcal{E} \longrightarrow \mathcal{M}_J = \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix} \quad (40)$$

with the following outputs:

$$\mathcal{D}(\text{order-parameter channel}) \mapsto J_H = 3/4, \quad (41)$$

$$\mathcal{D}(\text{adjoint/current channel}) \mapsto J_{\text{adj}} = 2, \quad (42)$$

$$\frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2} = \frac{x_+(J_H)}{x_+(J_{\text{adj}})} + \Delta_{\text{match}}. \quad (43)$$

The exact pole claim requires a derived vanishing of Δ_{match} or a computed matching remainder with stated scheme and source-theory origin. The theorem must also account for the negative branch in the same basis, so that the vector ratio and scalar clue arise from one operator.

The assignment target is a same-source projection package. A source variable u_{EW} must produce

$$u_{\text{EW}} \mapsto (\mathcal{H}_J, \langle \cdot, \cdot \rangle_{u_{\text{EW}}}, \Lambda_J, P_W, P_Z, P_\gamma, \mathcal{R}_{\text{pole}}, \mathcal{F}_{\text{sc}}), \quad (44)$$

where P_W is the charged pole sample, P_Z is the neutral massive pole sample, P_γ is the photon null sample, Λ_J fixes the dimensionless source normalization, $\mathcal{R}_{\text{pole}}$ is the source-to-complex-pole map, and $\mathcal{F}_{\text{sc}}$ is the scalar functional associated with the negative branch. The ordered electroweak assignment then has the target form

$$\begin{aligned} P_W(h_J) &= J_H = \frac{3}{4}, \\ P_Z(a_J^{\gamma^\perp}) &= J_{\text{adj}} = 2, \\ P_\gamma(a_\gamma) &= 0. \end{aligned} \quad (45)$$

The neutral entry requires a hypercharge normalization condition. Let $\mathcal{N}_T^\partial = \text{span}\{W_T^3, B_T\}$ be the transverse neutral trace space with the source product $\langle \cdot, \cdot \rangle_{u_{\text{EW}}}$. The standard neutral directions on this trace space are

$$a_\gamma^{(0)} = \frac{g'W_T^3 + gB_T}{\sqrt{g^2 + g'^2}}, \quad z^{(0)} = \frac{gW_T^3 - g'B_T}{\sqrt{g^2 + g'^2}}. \quad (46)$$

After projection to the current subspace, the photon trace is normalized in the source product:

$$\tilde{a}_\gamma = P_\gamma a_\gamma^{(0)}, \quad a_\gamma = \frac{\tilde{a}_\gamma}{\sqrt{\langle \tilde{a}_\gamma, \tilde{a}_\gamma \rangle_{u_{\text{EW}}}}}. \quad (47)$$

The massive neutral sample is

$$\Pi_{\gamma^\perp}^{(T)} = 1 - |a_\gamma\rangle\langle a_\gamma|_{u_{\text{EW}}}, \quad \tilde{a}_J^{\gamma^\perp} = P_{a,J}\Pi_{\gamma^\perp}^{(T)}z^{(0)}, \quad a_J^{\gamma^\perp} = \frac{\tilde{a}_J^{\gamma^\perp}}{\sqrt{\langle \tilde{a}_J^{\gamma^\perp}, \tilde{a}_J^{\gamma^\perp} \rangle_{u_{\text{EW}}}}}, \quad (48)$$

This definition carries admissibility conditions. The source product is required to be positive on the projected photon and Z traces,

$$\langle \tilde{a}_\gamma, \tilde{a}_\gamma \rangle_{u_{\text{EW}}} > 0, \quad \langle \tilde{a}_J^{\gamma^\perp}, \tilde{a}_J^{\gamma^\perp} \rangle_{u_{\text{EW}}} > 0. \quad (49)$$

The source theory must also state the ordered projection

$$P_{a,J}\Pi_{\gamma^\perp}^{(T)}$$

or prove a commutation relation for $P_{a,J}$ and $\Pi_{\gamma^\perp}^{(T)}$ on $\mathcal{N}_T^\partial$. These data belong to the same theorem package as

the hypercharge remainder below. The theorem data are

$$P_Z(a_J^{\gamma^\perp}) = 2 + \Delta_Y(u_{\text{EW}}), \quad P_\gamma(a_\gamma) = 0. \quad (50)$$

Here Δ_Y collects the hypercharge normalization, W^3/B mixing convention, source product, and photon-reference subtraction. The theorem target is a derived Δ_Y ; the special closure outcome is $\Delta_Y = 0$ in the selected convention. The corresponding pole package is

$$\frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2} = \frac{x_+(J_H)}{x_+(J_{\text{adj}})} + \Delta_{\text{O1}}(u_{\text{EW}}). \quad (51)$$

Here Δ_{O1} includes field normalization, threshold, scheme, and source-to-pole remainders. An exact pole statement is a theorem about Δ_{O1} .

Datum	Source-supported content	Target II theorem datum
h_J	Higgs doublet/order-parameter channel with $C_2(\mathbf{2}) = 3/4$	charged projector P_W from the same $\mathcal{H}_J$
$a_J^{\gamma^\perp}$	adjoint/current channel with $C_2(\text{adj}) = 2$	neutral massive projector P_Z after photon removal
a_γ	unbroken electro-magnetic direction	null projector P_γ and reference zero
Λ_J	shared dimension-less source scale	one normalization for J_H and J_{adj}
$\mathcal{R}_{\text{pole}}$	pole-scheme conversion map	complex-pole quotient with Δ_{O1}
u_-	negative-branch scalar channel	same-source scalar functional $\mathcal{F}_{\text{sc}}$

Coquereaux's algebraic $SU(2|1)$ superconnection gives source vocabulary in which Higgs fields enter generalized connection data [39]. Here that source supports the grading language for h_J and a_J . The ordered quotient in Eq. (51) requires the projectors, normalization, and pole remainder above.

Appendix C 5 records the interval-superconnection realization of this package. In that realization the CHM transverse Dirichlet-to-Neumann kernel, Coquereaux-type finite gauge-Higgs connection, photon subtraction, and source scale enter one reduced kernel $\hat{K}_J$. Its closure condition is the simultaneous derivation of $P_W(h_J) = 3/4$, $P_Z(a_J^{\gamma^\perp}) = 2$, $P_\gamma(a_\gamma) = 0$, and $\Delta_{O1/O10}$ in the complex-pole scheme.

A natural current to test is the Higgs isospin current

$$\mathcal{J}_\mu^a = iH^\dagger \frac{\sigma^a}{2} D_\mu H - i(D_\mu H)^\dagger \frac{\sigma^a}{2} H, \quad (52)$$

which transforms as an adjoint object and is built from the doublet order parameter. This current is a candidate bridge between the doublet source data and the adjoint current data. Turning this bridge into Eq. (43) is the remaining electroweak theorem target. In a general gauge, write

$$H = \begin{pmatrix} G^+ \\ (v + h + iG^0)/\sqrt{2} \end{pmatrix}.$$

The Higgs kinetic term contains the Goldstone-vector mixing terms

$$\mathcal{L}_{\partial GV} = iM_W [(\partial_\mu G^-)W^{+\mu} - (\partial_\mu G^+)W^{-\mu}] + M_Z(\partial_\mu G^0)Z^\mu. \quad (53)$$

This source-side expansion, reviewed in Ref. [9], is the equation-level test for the ordered current insertion: the charged insertion must satisfy

$$P_W(h_J) = \frac{3}{4} + \Delta_W(u_{EW}), \quad \Delta_W(u_{EW}) \stackrel{\text{closure}}{=} 0, \quad (54)$$

and the neutral insertion samples the photon-orthogonal current direction of Eqs. (48)–(50).

V. NEGATIVE BRANCH AND THE SCALAR SIDE OF THE PROBLEM

The negative branch in Eq. (6) is fixed by the same secular equation as the positive branch. The motivating datum is therefore the paired spectrum. Hans de Vries identified the positive-branch proximity to the mass-shell weak angle. Under the temporary normalization $M_Z^2 = \mu^2 x_+(2)$, the negative magnitudes define provisional scalar-function targets. That arithmetic observation supplies the main source-selection question: can one source theory attach $x_-(J)$ to a gauge-invariant scalar functional generated by the same reduced operator that gives $x_+(J)$?

The root identities imply

$$x_-(J) = -J - x_+(J) = -\frac{J}{x_+(J)}. \quad (55)$$

Any physical use of the positive branch automatically carries a partner branch.

The Higgs potential can be written as

$$V(H) = m^2 H^\dagger H + \lambda(H^\dagger H)^2, \quad m^2 < 0 \quad (56)$$

in the broken phase. The sign of the quadratic term makes the scalar interpretation a concrete target. If the normalization is fixed by $M_Z^2 = \mu^2 x_+(2)$, the two descriptive branch magnitudes are

$$M_{-,1} = M_Z \sqrt{\frac{|x_-(2)|}{x_+(2)}}, \quad M_{-,1/2} = M_Z \sqrt{\frac{|x_-(3/4)|}{x_+(2)}}. \quad (57)$$

Under this normalization, the two branch magnitudes are descriptive arithmetic. They become physical scalar identifications after the same determinant, normalization, selected J_* , and scheme are derived from gauge-Higgs or source-theory data.

A valid identification must be gauge-invariant and scheme-controlled. Possible targets include m^2 , λv^2 , the Higgs pole mass, a Wilson-line curvature, or an eigenvalue in a compactification problem. The branch values associated with $J = 3/4$ and $J = 2$ generate definite dimensionless numbers. The map from those numbers to scalar observables remains open.

A. Logical role of the negative root

The negative root changes the character of the clue. The full quadratic gives a paired object: a pole-ratio candidate and a scalar-side candidate arise from one secular equation. This pairing supplies the main conceptual reason to retain the negative branch. A dynamical theory that produces Eq. (2) has to say what both branches mean in the same basis.

The two-root structure can be stated as a spectral chain,

$$\mathcal{M}_J \longrightarrow \begin{cases} x_+(J) & \text{projective vector datum,} \\ x_-(J) & \text{candidate scalar/order-parameter target.} \end{cases} \quad (58)$$

The first line is the pole-ratio reading developed in Sec. III. The second line is a target for the Higgs, boundary, brane, or compactification sector. The chain is meaningful because both entries come from the same diagonalization. Any proposed derivation that keeps the positive branch acquires the negative-branch obligation automatically.

This obligation also sharpens the electroweak assignment. The ordered pair $(J_H, J_{\text{adj}}) = (3/4, 2)$ already suggests two electroweak structures: the doublet order parameter and the adjoint current. The negative branch asks whether the doublet/order-parameter side can be read as a scalar instability, a vacuum scale, or a deformation mode in the same operator that gives the vector pole quotient. The scalar reading therefore ties together three open calculations: the ordered J -assignment, the determinant derivation, and the branch-to-Higgs map.

B. Gauge-invariant scalar language

The scalar sector has several layers. The Higgs doublet H itself is gauge-covariant. The radial composite

$$\varphi^2 = 2H^\dagger H \quad (59)$$

is the local invariant order-parameter language used here after renormalization and scheme choice. In the broken phase, a gauge choice may represent the vacuum by a real radial expectation value, yet the physical statement should be expressible through invariant quantities, pole positions, or matched parameters. The negative branch therefore requires a defined scalar functional before it can carry a physical claim.

An abstract target is

$$\mathcal{F}_{\text{sc}} = \mathcal{F}_{\text{sc}}(H^\dagger H, (D_\mu H)^\dagger D^\mu H, V(H), \text{boundary or compactification data}), \quad (60)$$

with a branch assignment

$$\mathcal{F}_{\text{sc}}(J_\star; \Lambda, s_{\text{ren}}, u) = C_{\text{sc}}(\Lambda, s_{\text{ren}}, u) |x_-(J_\star)|. \quad (61)$$

Here $J_\star$, C_{sc} , Λ , the renormalization scheme s_{ren} , and the source datum u are part of the theorem. The expression may describe a pole mass, a curvature of the scalar potential, a vacuum-ray coordinate, a brane separation, a Wilson-line curvature, or a compactification deformation. These readings remain candidate targets until one route derives a specific object.

The same-source package needed for a physical reading is

$$u \longmapsto (\mathcal{H}_J, \langle \cdot, \cdot \rangle_u, P_J, \Lambda_J, \mathcal{R}_{\text{pole}}, \mathcal{F}_{\text{sc}}), \quad (62)$$

with $\mathcal{H}_J$ the reduced two-channel space, P_J the projection, Λ_J the source scale, and $\mathcal{R}_{\text{pole}}$ the pole map used for the vector branch. A dimensionless scalar target then has the form

$$\widehat{\mathcal{F}}_{\text{sc}}(u; J_\star) = \Lambda_J^{-d_{\text{sc}}} \mathcal{F}_{\text{sc}}(J_\star; \Lambda_J, s_{\text{ren}}, u) = C_{\text{sc}}(s_{\text{ren}}, u) |x_-(J_\star)|, \quad (63)$$

where d_{sc} is the mass dimension of the chosen observable. The source derivation must state whether $\widehat{\mathcal{F}}_{\text{sc}}$ is a vacuum-ray coordinate, a potential curvature, a pole observable, a boundary modulus, or a compactification eigenvalue.

The target can be summarized by the commutative source diagram

$$\begin{array}{ccc} \mathcal{L}_{\text{src}}(u) & \longrightarrow & (\mathcal{H}_J, \langle \cdot, \cdot \rangle_u, P_J) \longrightarrow & Q(J) \\ & & & \downarrow \\ & & & \begin{cases} x_+(J) & \text{vector pole quotient,} \\ x_-(J) & \text{candidate scalar funct} \\ x_-(J) & \text{auxiliary or gauge-fixe} \end{cases} \end{array} \quad (64)$$

The third line is the explicit failure outcome for a scalar interpretation.

A sharper downstream target appears if the scalar functional is the vacuum-ray normalization. In that case the Standard Model identity $M_W^2 = g^2 v^2 / 4$ can turn the pair $x_+(3/4)$ and $x_-(2)$ into a candidate construction coupling. Appendix C 4 records this as an electromagnetic endpoint theorem target: the scalar branch, the charged-vector normalization, the $D = 9$ $U(1)_{\text{em}}$ kinetic term, and the matching pair $(Q_\alpha, \Delta_{\alpha, \text{match}})$ must be derived before α acquires construction status.

The scalar pole target is the most restrictive because it demands the same pole discipline as the vector comparison. The potential target is more sensitive to matching conventions, because m^2 , λ , and v depend on the renormalization scheme and on how the effective potential is defined. The compactification target is broader: $x_-(J)$ may first appear as an eigenvalue or deformation parameter in a higher-dimensional problem, with the four-dimensional Higgs interpretation emerging after reduction.

C. Vacuum ray and branch pairing

The electroweak ray separates scale from direction. Along the broken-to-unbroken ray, the radial coordinate

scales the vector masses, while the ratio

$$\frac{M_W^2}{M_Z^2} = \frac{g^2}{g^2 + g'^2} \quad (65)$$

is projective at tree level. The DeVries quotient addresses this projective datum. The negative branch is a scalar-side target because it shares the same two-channel operator while carrying the opposite eigenvalue.

In a route with fields h_J and a_J , the branch vectors have the schematic form

$$u_{\pm}(J) = \alpha_{\pm}(J)h_J + \beta_{\pm}(J)a_J. \quad (66)$$

The scalar-side reading becomes a statement about the h_J component of $u_{-}(J)$. The vector-side reading becomes a statement about the projective pole ratio extracted from $u_{+}(J)$. A source derivation must therefore give a branch basis, an inner product, and a normalization. These data determine whether the negative eigenvector genuinely selects an order-parameter amplitude or whether it belongs to an auxiliary sector of the chosen source theory.

The electroweak ray also supplies a consistency check. Taking the radial vacuum coordinate toward the symmetric point should collapse the vector masses together with the Higgs order parameter. The DeVries ratio can remain as a projective direction in coupling or spectral space. A scalar interpretation of x_{-} should respect this ray structure by scaling with the radial data or by controlling the deformation that sets the radial scale.

D. Candidate scalar targets

The Standard Model scalar sector gives several inequivalent objects that can carry a scalar-side interpretation [8, 9]. They have different scheme and gauge properties:

Target	Meaning	DeVries requirement
$m^2(s_{\text{ren}}, \Lambda)$	quadratic term in the Higgs potential	define the scheme and invariant extraction
$\lambda(s_{\text{ren}}, \Lambda)v^2$	curvature scale along the radial mode	connect branch value to the electroweak ray
$M_{h,\text{pole}}^2$	scalar pole position	derive the scalar pole from the same determinant as W/Z
$\partial_{\alpha}^2 V_{\text{eff}}(\alpha) _{\alpha_{\star}}$	Wilson-line Hosotani curvature	or identify the holonomy and source normalization
radion/modulus eigenvalue	compactification scalar or boundary deformation	identify the compact operator and its normalization
brane separation mode	endpoint or brane order parameter	show coupling to the adjoint/current channel

The pole interpretation is the sharpest target. It asks the same object that governs the vector pole quotient to generate a scalar pole or scalar instability. The potential interpretation is more flexible, because m^2 , λ , and v depend on conventions and matching. The compactification interpretation is broader still; it may identify the negative branch with a modulus or boundary deformation whose four-dimensional scalar meaning emerges after reduction.

E. Route-specific scalar maps

The endpoint route maps the negative branch to brane scalar data. In a D-brane system, transverse positions of branes appear as scalar fields on the worldvolume, while strings stretched between branes acquire masses controlled by separation data [2, 3]. A DeVries endpoint derivation would identify $u_{-}(J)$ with a brane-Higgsing direction, a stretched-string scalar, or a boundary order-parameter field. The map has to use the same endpoint algebra that supplies the current trace and off-diagonal product.

The interval route maps the negative branch to boundary or modulus data. A five-dimensional gauge field supplies four-dimensional vector modes and scalar components; the compactification radius, brane terms, and boundary conditions determine the light spectrum [7, 16]. A DeVries interval derivation would identify $u_{-}(J)$ with a boundary scalar amplitude, an A_5 -like scalar, a radion-like deformation, or a brane-localized order parameter. The map has to preserve the same boundary determinant that gives the W/Z branch quotient. The Hosotani version of this target uses the holonomy variable

$$\alpha_H \sim g_5 \int_I dy A_5, \quad \mathcal{F}_{\text{sc}}^{\text{Hol}} = \left. \frac{\partial^2 V_{\text{eff}}(\alpha_H; \Lambda, s_{\text{ren}})}{\partial \alpha_H^2} \right|_{\alpha_H = \alpha_{\star}}, \quad (67)$$

and asks whether the same interval kernel that yields the transverse vector pole also yields $\hat{\mathcal{F}}_{\text{sc}}^{\text{Hol}} = C_{\text{sc}}|x_{-}(J_{\star})|$ [29, 30, 34].

The G_2 route maps the negative branch to singularity deformation data. Local G_2 constructions place gauge sectors and charged matter at singular loci, and Higgs-bundle descriptions encode matter localization and interaction data through functions on an associative three-cycle [18–20]. A DeVries G_2 derivation would identify $u_{-}(J)$ with a deformation parameter, a localized modulus, a critical value of f_Q , or a charge-density functional. The same local geometry then has to produce the adjoint/current and order-parameter entries of the kernel.

The route comparison is:

Route	Source-supported scalar variable	Missing DeVries map
Endpoint	transverse scalar, string separation order parameter	brane derive the same endpoint determinant, projection, and normalization for $u_-(J)$
Interval	boundary Higgs datum, A_5 holonomy, boundary condition modulus	derive the same interval kernel, gauge fixing, scalar/eaten separation, and Λ_J
G_2	singularity deformation, Higgs-bundle one-form, localized modulus	derive the local two-channel kernel and the scalar functional on the same associative-cycle data

These maps give the negative branch a practical role in route selection. A candidate route that produces the positive-branch quotient and gives the negative eigenvector a scalar or deformation interpretation gains additional physical content. A route that assigns the negative branch to an auxiliary variable defines a narrower physical claim.

F. Eigenvectors and branch meaning

The two roots also have eigenvectors. For the matrix in Eq. (4), an eigenvector with eigenvalue x can be represented by

$$v_x \propto \begin{pmatrix} \sqrt{J} \\ x \end{pmatrix}. \quad (68)$$

Thus the branch assignment is a statement about channel composition. The positive and negative roots are mixtures of the same two basis amplitudes. A scalar interpretation of x_- should therefore identify the basis in which one component is the vector/current side and the other is the order-parameter or boundary side.

A criterion is the following. If the same two-channel determinant produces both the vector pole ratio and the scalar instability, then the determinant should appear in an effective action, boundary condition, or self-energy equation. The corresponding theorem target is an explicit determinant with an identified second-channel field or mode. Breitenlohner–Freedman stability supplies a route caveat for AdS scalar readings: a negative mass-squared can be admissible under specified boundary conditions, while the DeVries sign still requires the source functional in Eq. (63) [35].

VI. STRING, KALUZA–KLEIN, AND G_2 COMPLETION PROBLEM

The secular equation has the form of a low-level branch splitting that preserves a high-spin datum. This is the

reason string and Regge language enter the construction. The original Rivero construction imposed a high-spin condition motivated by Regge behavior [1]. A string interpretation must therefore preserve the asymptotic slope while explaining why the low levels split through the two-channel determinant.

The string-theory input is concrete. Quantized strings generate towers, level matching, endpoint sectors, winding, and compactification charges. A compact circle already contains the basic Kaluza–Klein mechanism: a higher-dimensional metric reduces to a lower-dimensional metric, a gauge field, and a radius modulus; Fourier modes along the circle acquire masses n^2/R^2 and charges under the gauge field descending from the metric [2]. The string worldsheet adds winding sectors, and winding modes couple to the vector descending from the antisymmetric tensor. Compactification therefore supplies more than one integer or representation label, and different sectors can couple to different gauge fields.

Open strings add boundary data. Neumann and Dirichlet boundary conditions relate left- and right-moving oscillators differently, and D-branes localize open-string states on submanifolds [2]. Coincident branes generate nonabelian gauge fields from open-string excitations. A DeVries derivation based on endpoints or branes must identify the two channels in Eq. (69) with actual boundary degrees of freedom, endpoint charges, or brane/bulk mixing modes.

The older phenomenological string lineage is relevant here. Regge trajectories in hadron spectra motivate a rotating flux-tube picture, and the open-string Veneziano amplitude was first written to capture strong-interaction features before QCD supplied the gauge-theory description [2]. In this lineage, a low-energy spectral regularity can guide the search for a deeper string or boundary structure, provided the proposed determinant is eventually derived.

Endpoint physics gives the simplest analogy for charged vector states. In a coincident-brane system, strings stretching between branes carry off-diagonal gauge quantum numbers, and the massless excitations organize into a nonabelian gauge field [2]. The electroweak target is a broken nonabelian gauge sector coupled to a doublet order parameter. A successful endpoint model must therefore produce both the adjoint-current datum $J = 2$ and the doublet/order-parameter datum $J = 3/4$ inside one boundary or brane determinant.

A Kaluza–Klein or brane derivation would have to produce the determinant

$$\det \begin{pmatrix} x & -\sqrt{J} \\ -\sqrt{J} & x + J \end{pmatrix} = 0. \quad (69)$$

This can arise from a two-channel boundary problem, a mixing of brane-localized and bulk modes, or an effective orbit problem. The required output is the exact J -dependence in Eq. (69).

Extra-dimensional electroweak models show that boundary conditions can break gauge symmetry and de-

termine vector spectra. Higgsless constructions on intervals provide a controlled language for relating boundary data, vector masses, fermion localization, and precision observables [7]. The interval viewpoint is valuable because boundary conditions can be obtained as limits of boundary dynamics. For the present construction, this gives a precise target: derive the $J = 3/4, 2$ samples as boundary data associated with the Higgs doublet/order-parameter channel and the adjoint/current channel.

Witten's search for realistic Kaluza–Klein theory gives the historical benchmark. Extra dimensions can generate gauge groups from geometry, including $SU(3) \times SU(2) \times U(1)$, while the realistic fermion quantum numbers impose severe constraints [16]. The DeVries construction inherits this lesson. A compactification mechanism must solve the electroweak spectral problem and remain compatible with chiral matter.

M-theory on G_2 -holonomy manifolds supplies a separate completion map. Smooth compact G_2 compactifications give four-dimensional minimal supersymmetry in the large smooth limit. Realistic charged chiral matter requires singularities supporting nonabelian gauge sectors and chiral fermions [17–19]. Local Higgs-bundle descriptions over associative three-cycles give a modern way to organize gauge sectors, localized matter, and singular transitions [20].

The completion problem can therefore be stated as a list of mechanisms and obligations.

Mechanism	What it can supply	Required DeVries test
Regge/string tower	high-spin totics	preserve slope and produce Eq. (69)
Circle KK reduction	gauge fields and charged KK modes	identify the charge/Casimir source of J
Winding sector	second compactification charge	test a winding-like interpretation of the second branch
Interval/brane BC	vector spectra and boundary dynamics	derive $J = 3/4, 2$ samples from boundary data
Endpoint/open-string sector	localized gauge states	identify the two channels as endpoint or brane/bulk modes
Singular G_2	gauge/chiral sectors	localize the electroweak operator geometrically
Higgs bundle	matter/mass localization	identify the two branches as spectral-cover data

VII. RADICAL PLACEMENT AND SCALE ATTACHMENT

The secular equation contains the radical

$$\Delta_J^{1/2} = \sqrt{J^2 + 4J}. \quad (70)$$

This paper treats $\Delta_J^{1/2}$ as the central object whose physical address must be identified. The algebra alone specifies a two-branch spectrum. Physics begins when the radical is assigned to a pole condition, a boundary determinant, a compactification eigenvalue, a worldsheet constraint, or a running matching condition.

A. Placement options

The possible placements are distinct assignments of the same radical.

Placement	Physical meaning	Required derivation
Pole spectrum	$\Delta_J^{1/2}$ enters propagator self-energy condition	derive the W/Z pole ratio and scalar partner branch from one spectral equation
GUT boundary	$\Delta_J^{1/2}$ fixes a high-scale weak-angle datum	embed the J -samples in a unification representation and run to the pole scale
Running parameter	$\Delta_J^{1/2}$ constrains a scheme-dependent coupling or vacuum parameter	specify the scheme, scale, matching map, and threshold corrections
KK eigenvalue	$\Delta_J^{1/2}$ comes from a compact internal wave operator	identify the internal operator, mode labels, denal wave operator generacies, and boundary conditions
Open-string endpoint	$\Delta_J^{1/2}$ comes from endpoint or Chan–Paton data	derive the two-channel matrix from endpoint charges and brane/bulk mixing
Electroweak superconnection	$\Delta_J^{1/2}$ is attached to odd Higgs data and even transverse-current data in a generalized gauge-Higgs connection	derive $P_W : \Phi_{\text{odd}} \mapsto 3/4$, $P_Z : F_{\text{even}}^{\gamma^\perp} \mapsto 2$, and the pole matching from one complex
Worldsheet constraint	$\Delta_J^{1/2}$ comes from level matching, conformal to the electroweak determinants, or a modular sector	identify the CFT labels and show how they reduce to the electroweak determinant

The pole-spectrum placement is the hypothesis used below. It gives the numerical clue its electroweak content because W and Z are observed as poles. It also gives the negative branch a task: the same determinant should know about the scalar/order-parameter side. This reading demands the most direct dynamical calculation, since it asks string/Kaluza–Klein structure to survive down to a low-energy pole observable.

The GUT-boundary placement gives a familiar comparison class. In $SU(5)$ and $\text{Spin}(10)$ settings, representation theory can give a clean weak-angle value at the

high scale, with running and thresholds supplying the low-energy relation [24]. A DeVries high-scale reading would join that lineage. This paper pursues the pole reading because the negative branch and pole-ratio clue appear together in the same low-energy normalization.

B. Scale attachment

Let $\mathcal{O}_J$ denote the physical operator whose two-channel reduction gives

$$\det(\lambda - \mathcal{O}_J) = \lambda^2 + J\lambda - J. \quad (71)$$

The scale question is then a question about where $\mathcal{O}_J$ lives:

$$\mathcal{O}_J \in \{\mathcal{O}_{\text{pole}}, \mathcal{O}_{\text{GUT}}, \mathcal{O}_{\text{KK}}, \mathcal{O}_{\text{brane}}, \mathcal{O}_{\text{CFT}}\}. \quad (72)$$

Each address has a different interpretation of the same algebra. A pole operator must be built from four-dimensional gauge-Higgs correlators. A GUT operator must be built from representation traces and matching. A KK operator must be a compact internal differential operator or boundary value problem. A brane operator must couple localized and bulk modes. A worldsheet operator must arise from conformal weights, current algebra, or level matching.

This list fixes the scheme dependence of the radical. A different placement changes the required derivation.

C. Placement chain

The placement question can be recorded as a chain. A source theory $\mathcal{S}$ supplies an operator, the operator supplies a two-channel kernel, the kernel supplies branches, and the branches acquire physical meaning after a scheme and an assignment rule have been fixed:

$$\begin{aligned} \mathcal{S} &\longrightarrow \mathcal{O}_J \longrightarrow \mathcal{M}_J \longrightarrow \{x_+(J), x_-(J)\}, \\ \{x_+(J), x_-(J)\} &\longrightarrow \{s_{\text{pole}}^2, \mathcal{F}_{\text{sc}}, \mathcal{C}_{\text{top}}\}. \end{aligned} \quad (73)$$

Here $\mathcal{F}_{\text{sc}}$ denotes the scalar/order-parameter functional sought in Sec. V, and $\mathcal{C}_{\text{top}}$ denotes the global-form and representation constraints discussed in Secs. XIV and XV. The chain separates four tasks:

$$\text{operator origin: } \mathcal{S} \mapsto \mathcal{O}_J, \quad (74)$$

$$\text{two-channel reduction: } \mathcal{O}_J \mapsto \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}, \quad (75)$$

$$\text{branch reading: } \mathcal{M}_J \mapsto \{x_+(J), x_-(J)\}, \quad (76)$$

$$\text{physical assignment: } \{x_{\pm}\} \mapsto (s_{\text{pole}}^2, \mathcal{F}_{\text{sc}}, \mathcal{C}_{\text{top}}). \quad (77)$$

Each arrow has different source support. Complex-pole and scheme language belongs to the last arrow [10].

Higgs doublet, adjoint current, custodial structure, and the electroweak ray belong to the assignment arrow [8, 9]. Endpoint labels, D-brane gauge fields, and interval boundary conditions belong to the first two arrows [2, 3, 7]. Equation (76) is algebraically controlled; the other arrows are open derivation targets.

This chain also organizes the negative branch. If a candidate source theory derives $\mathcal{M}_J$, then $x_-(J)$ is carried by the same light-sector basis as $x_+(J)$. The scalar question becomes a question about the eigenvector map,

$$u_-(J) \in \mathcal{H}_J \longmapsto \mathcal{F}_{\text{sc}}, \quad (78)$$

where $\mathcal{H}_J$ is the two-dimensional light sector. A Higgs interpretation requires $\mathcal{F}_{\text{sc}}$ to be gauge invariant. A brane interpretation requires $\mathcal{F}_{\text{sc}}$ to be a modulus, separation, or localized boundary datum. A compactification interpretation requires $\mathcal{F}_{\text{sc}}$ to be a normalized internal eigenmode or deformation parameter.

The chain gives a compact criterion for acceptance. A proposed mechanism should fill one of the arrows in Eq. (73) or state a new arrow with its source theory, scheme, and physical assignment.

D. Original research route

The route developed here is:

$$\begin{aligned} \text{Regge/high-spin datum} &\longrightarrow \text{two-channel secular determinant} \\ &\longrightarrow \text{electroweak pole ratio and scalar part} \end{aligned} \quad (79)$$

The first arrow belongs to the string/dual-model tradition: spectra organize into towers with controlled asymptotics. The second arrow is the DeVries algebra. The third arrow is the remaining electroweak derivation.

A concrete model should therefore supply four ingredients:

1. an internal label whose quadratic invariant is J ;
2. a two-channel system whose off-diagonal mixing is fixed by $\sqrt{J}$;
3. a diagonal trace term giving the $-J$ entry;
4. a physical normalization that places the positive branch on the vector pole ratio and the negative branch on the scalar/order-parameter side.

The fourth ingredient is the new conceptual burden. It turns the algebraic clue into a physical claim hierarchy and fixes the relation between the pole reading and high-scale weak-angle boundary data.

VIII. DUAL-MODEL LINEAGE AND OPEN-STRING ENDPOINT DATA

The DeVries construction enters this paper through a spectral regularity. This is why the older dual-model lineage matters. The Veneziano amplitude was introduced

to capture strong-interaction regularities before its string interpretation became clear; its open-string form later supplied a world-sheet explanation of the pole tower and the associated Regge behavior [2]. The lesson for the present work is methodological and technical. A compact algebraic regularity in particle data can guide a deeper spectral construction when the amplitude or boundary problem subsequently explains the regularity.

The relevant feature is the coexistence of a low-level spectrum and a high-spin asymptotic rule. In string amplitudes, pole residues organize the spins carried by the exchanged string states, and the tower structure follows from the quantized oscillators [2]. For the DeVries determinant, the secular equation must play an analogous role at a much lower and more constrained level:

$$\text{high-spin/Regge regularity} \rightsquigarrow x^2 + Jx - J = 0 \rightsquigarrow \{x_+(J), x_-(J)\} \quad (80)$$

The first arrow is historical motivation until a world-sheet or boundary calculation produces the second expression. The second arrow is algebraic and already fixed. The Rivero/DeVries observation functions as a clue that selects a target determinant.

The same distinction controls the higher positive branch. Ordinary open-string Regge behavior is supplied by an oscillator level N_{osc} and a common slope; the DeVries label has to be a sector or intercept label j , with $A_j = j(j+1)$, if higher slots are to coexist with a straight trajectory. The conditional tower has the form

$$M_{N_{\text{osc}}, j, \sigma}^2 = \mu^2 x_\sigma(A_j) + \frac{N_{\text{osc}}}{\alpha'} + \Delta_{N_{\text{osc}}, j, \sigma}^{\text{Regge}}, \quad \sigma = \pm. \quad (81)$$

Here j labels the reduced DeVries sector, while N_{osc} and the trajectory spin label the string excitation. The $j = 3/2$ positive slot is therefore a branch-assignment target. Particle identity, gauge quantum numbers, production, decay, and collider compatibility must come from the same source theory that derives the determinant.

A related brane-scaling test adds an axis map to Eq. (81). A source theory must state whether the DeVries sector label maps to a putative rotating-brane angular momentum or serves as an intercept label. The shared source package is

$$\mathfrak{R}_j = (K_j, \alpha', P_{\text{surv}}, \mathcal{B}_{\text{ND}}, \mathcal{Q}, \mathcal{T}, \chi_{\text{spin}}),$$

where χ_{spin} is the extra axis map. The current algebraic test is summarized in Target C 5: the positive branch is bounded at large putative brane spin, while the negative magnitude grows linearly under the mass reading.

Open strings provide a more concrete mechanism than closed-string Regge language alone. An open-string state carries oscillator data in the interior and boundary data at its endpoints. D-branes give a geometric home to the endpoints, and coincident branes give nonabelian gauge fields from open-string modes [2, 3]. This suggests a precise kind of model for Eq. (69): one channel should be an adjoint or current channel associated with gauge fields,

while the other should carry the order-parameter data that samples the doublet invariant.

For the DeVries determinant, the endpoint route can be written as a quadratic action for two boundary-coupled amplitudes,

$$S_J^{(2)} = (\psi_1^\dagger \ \psi_2^\dagger) \begin{pmatrix} \lambda & -\kappa_J \\ -\kappa_J & \lambda + \tau_J \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}, \quad (82)$$

with the DeVries target

$$\kappa_J^2 = J, \quad \tau_J = J. \quad (83)$$

Here ψ_1, ψ_2 are placeholders for actual endpoint, brane, or localized modes. The notation is intentionally sparse. The physics is carried by the derivation of κ_J and τ_J .

There are three natural endpoint readings.

1. *Chan-Paton/current reading.* The endpoint labels carry gauge indices. The diagonal entry τ_J is a trace or current-algebra invariant, and the off-diagonal entry κ_J is a group-theoretic transition matrix element.
2. *Brane/bulk reading.* One amplitude is localized on a brane or boundary, while the other propagates in the bulk. The determinant comes from integrating over the bulk mode and imposing the boundary matching condition.
3. *Order-parameter reading.* One channel is an adjoint-current datum and the other is a doublet/order-parameter datum. This is the route that can explain why the electroweak samples are $J = 2$ and $J = 3/4$.

Each reading must respect the same electroweak constraint. The pole ratio compares charged and neutral vector poles, while the $J = 3/4$ datum belongs naturally to the Higgs doublet side. A successful endpoint construction therefore has to turn an order-parameter doublet invariant into the charged-vector sample in the pole quotient. This is the ordered-sampling problem in boundary language.

The open-string route also clarifies the negative branch. The two roots are eigenvalues of one two-channel system. If the positive branch is associated with the vector pole quotient, the negative branch should correspond to the partner channel in the same boundary problem. In an endpoint construction this partner may be a scalar boundary modulus, a Higgs-like order parameter, a brane separation mode, or an auxiliary channel eliminated in the low-energy action. This makes the negative branch a demand on the model: the same determinant has to explain both eigenvalues.

This places the string interpretation in a concrete form. The target is an endpoint or brane calculation whose reduced determinant has the entries in Eq. (83), whose low-energy normalization gives the W/Z pole quotient, and whose partner eigenvalue lands in the scalar/order-parameter sector.

IX. KALUZA–KLEIN SPECTRA, BOUNDARY CONDITIONS, AND SECULAR DETERMINANTS

Kaluza–Klein theory supplies the second concrete language for the DeVries program. The original idea unifies gauge fields with geometry by expanding a higher-dimensional theory around a compact ground state. In Witten’s account, symmetries of the compact space appear as gauge symmetries in the lower-dimensional theory, and the physical spectrum is obtained from small oscillations around the compactification background [16]. Salam and Strathdee give the complementary harmonic-expansion language on compact homogeneous spaces G/H , where zero modes give gravitational plus Yang–Mills fields and higher modes give massive multiplets [33]. The same account also states the hard model-building burden: obtaining the Standard Model gauge group, fermion quantum numbers, and scalar sector is a serious representation problem, even when the gauge group itself can be engineered.

The DeVries problem asks for a narrower and sharper object inside this tradition: a compact or interval derivation of the two-channel determinant

$$\det \begin{pmatrix} \lambda & -\sqrt{J} \\ -\sqrt{J} & \lambda + J \end{pmatrix} = \lambda^2 + J\lambda - J. \quad (84)$$

This determinant is the entire target of the KK route.

A. Circle compactification and two charge channels

String compactification on a circle gives a clean source-backed example of a two-channel spectral structure. Ordinary Kaluza–Klein momentum is quantized along the circle. Closed strings add winding around the same circle. Polchinski’s circle example shows that the Kaluza–Klein vector couples to compact momentum, while the vector descending from the antisymmetric tensor couples to winding [3]. At the self-dual radius the spectrum has enhanced current algebra, and changing the radius is described by an operator built from the same currents.

This gives a source-backed analogy:

$$\begin{pmatrix} \text{momentum-like channel} \\ \text{winding-like or boundary channel} \end{pmatrix} \rightsquigarrow \{x_+(J), x_-(J)\} \quad (85)$$

The analogy becomes a derivation after the two channels are reduced to Eq. (84). The self-dual-radius example also explains why a scalar modulus belongs in the discussion. The compact radius is a scalar field in the lower-dimensional theory, and gauge enhancement can occur at special values of that scalar. This is the compactification analogue of the positive/negative branch problem: a vector spectral statement and a scalar modulus can arise from the same compact sector.

A sharper test combines this channel picture with a boundary Schur complement. Let the compact or interval

sector contain a momentum/current-like channel and a winding, brane-separation, or order-parameter channel. After the tower is integrated out, the reduced boundary kernel should have the form

$$K_j^{\text{eff}}(\lambda) = K_j^{\text{bdry}}(\lambda) - V_j^\dagger (\lambda - L_j)^{-1} V_j \longrightarrow \begin{pmatrix} \lambda & -\sqrt{A_j} \\ -\sqrt{A_j} & \lambda + A_j \end{pmatrix}, \quad A_j \quad (86)$$

This target is the KK form of the Regge-intercept reading in Sec. VIII: the compact or boundary calculation supplies the two-channel intercept, and the oscillator or tower label supplies the large-spin growth.

B. Interval variational data

Extra-dimensional field theory on an interval gives a more direct determinant language. Csaki–Hubisz–Meade derive boundary conditions from the variational principle for fields on an interval and show how boundary mass terms and boundary kinetic terms modify those conditions [7]. Dirichlet–Higgs and radion-stabilization models give adjacent source variables for boundary scalar profiles, radion data, and bulk Higgs stabilizers [29, 30]. A scalar example has the schematic action

$$S_{\text{int}} = \int d^4x \int_0^L dy \frac{1}{2} [\partial_M \Phi \partial^M \Phi - V(\Phi)] + S_0[\Phi] + S_L[\Phi], \quad (87)$$

with boundary functionals S_0 and S_L . The boundary variation supplies equations at $y = 0$ and $y = L$. Boundary mass terms give Robin-type conditions. Boundary kinetic terms can make the boundary equation depend on the four-dimensional eigenvalue. This is exactly the structural feature needed by a secular determinant.

For a mode $\Phi(x, y) = \phi_\lambda(x) f_\lambda(y)$, the interval problem has the abstract form

$$\mathcal{L}_J f_\lambda(y) = \lambda f_\lambda(y), \quad 0 < y < L, \quad (88)$$

$$\mathcal{B}_{0,J}(\lambda) f_\lambda(0) = 0, \quad \mathcal{B}_{L,J}(\lambda) f_\lambda(L) = 0. \quad (89)$$

The DeVries route asks for a two-dimensional light sector $\mathcal{H}_J$ extracted from this problem. A convenient basis is

$$\mathcal{H}_J = \text{span}\{h_J, a_J\}, \quad (90)$$

where h_J is a boundary order-parameter amplitude and a_J is an adjoint/current amplitude. The interval derivation then reduces the bulk-plus-boundary problem to an effective kernel

$$K_J^{\text{int}}(\lambda) = \begin{pmatrix} \lambda + \Sigma_{hh,J}^{\text{int}}(\lambda) & \Sigma_{ha,J}^{\text{int}}(\lambda) \\ \Sigma_{ah,J}^{\text{int}}(\lambda) & \lambda + \Sigma_{aa,J}^{\text{int}}(\lambda) \end{pmatrix}. \quad (91)$$

The DeVries target is the boundary limit

$$\Sigma_{hh,J}^{\text{int}} = 0, \quad \Sigma_{aa,J}^{\text{int}} = J, \quad \Sigma_{ha,J}^{\text{int}} \Sigma_{ah,J}^{\text{int}} = J. \quad (92)$$

This is the interval version of the endpoint target in Sec. XI.

CHM boundary data supply the interval version of this target. In the scalar example, a boundary kinetic term at $y = 0$ has the source normalization

$$S_{\text{brane,kin}}^{(0)} = \int d^4x \frac{1}{2M_0} \partial_\mu \phi \partial^\mu \phi|_{y=0}, \quad \partial_y f_n(0) = -\frac{m_n^2}{M_0} f_n(0) \quad (93)$$

The associated product is

$$(f, g)_{\text{CHM}} = \int_0^{\pi R} f(y)g(y) dy + \sum_i \frac{1}{M_i} f(y_i)g(y_i). \quad (94)$$

For gauge fields, boundary scalar vevs give vector boundary data of the source form

$$\partial_y A_\mu|_{y_i} \mp v_i^2 A_\mu(y_i) = 0, \quad \left(\frac{m^2}{\xi_i} - v_i^2 \right) \pi_i + v_i A_5|_{y_i} = 0, \quad (95)$$

with endpoint-orientation signs fixed by the variational convention. The CHM higgsless/custodial construction also supplies the photon zero-mode and custodial boundary accounting used by the later electroweak boundary proof. These source facts define the allowed boundary-kernel class. The equality to Eq. (92) remains the open DeVries entry target. The scalar equation makes the gauge-fixing burden explicit: the projected DeVries kernel must state the treatment of ξ_i , separate physical scalar data from eaten Goldstone data, preserve the photon zero mode, and map the interval eigenvalue into the four-dimensional transverse pole condition.

A Hodge factorization subroute gives a sharper test for the off-diagonal entry. Witten's supersymmetric quantum mechanics on forms uses the exterior derivative and its adjoint to build supercharges whose Hamiltonian is the Hodge Laplacian; the Witten deformation adds Morse-function data to the same complex [36]. If an interval or compact sector supplies normalized states $e_J^0 \in \Omega^0$ and $e_J^1 \in \Omega^1$ with

$$D_J e_J^0 = \sqrt{J} e_J^1, \quad D_J^\dagger e_J^1 = \sqrt{J} e_J^0, \quad (96)$$

then the de Rham part of the reduced operator supplies $\Sigma_{ha,J}^{\text{int}} \Sigma_{ah,J}^{\text{int}} = J$. In Target X language, Eq. (96) supplies $Q_{\text{dR},J}$; the boundary or breaking datum B_J must supply the diagonal entry $-J$. The same interval model must derive the sign convention in the DeVries block and the ordered electroweak map. Appendix C5 records this as a theorem target.

This gives the interval route an entry order. The Hodge/SUSY-QM calculation tests the square-root product $\Sigma_{ha,J}^{\text{int}} \Sigma_{ah,J}^{\text{int}} = J$. The CHM boundary calculation tests the diagonal current entry $\Sigma_{aa,J}^{\text{int}} = J$. A successful route must place both tests in one source Hilbert space, with one inner product, one projection to $\mathcal{H}_J$, and one pole-matching map. The negative branch is then read from the eigenbasis of the same reduced kernel.

C. Dimensional Schur-complement target

The dimensional interpolation idea defines a candidate Schur-complement calculation within Target III. Let the middle-dimensional sector be split into a light boundary subspace $\mathcal{H}_J = \text{span}\{h_J, a_J\}$ and a tower, bulk, or compact sector with operator L_J . With a boundary kernel $K_J^{\text{bdry}}(\lambda)$, coupling map V_J , and specified inner product, integrating out the heavy sector gives

$$K_J^{\text{eff}}(\lambda) = K_J^{\text{bdry}}(\lambda) - V_J^\dagger (\lambda - L_J)^{-1} V_J. \quad (97)$$

The required reduced kernel is

$$K_J^{\text{eff}}(\lambda) \longrightarrow \begin{pmatrix} \lambda & -\sqrt{J} \\ -\sqrt{J} & \lambda + J \end{pmatrix} \quad (98)$$

in a limit where additional light channels decouple or acquire a controlled block-diagonal form. This formulation connects the $D = 10/6$ middle line to the route kernel: K_J^{bdry} contains boundary kinetic and mass data, L_J contains the integrated tower, and V_J contains the order-parameter/current mixing. A successful interval derivation must specify these three objects, the inner product, and the normalization yielding Eq. (92).

The same target can be stated in Dirichlet-to-Neumann form. Solving the bulk mode equation with fixed boundary values gives a map from boundary traces to normal derivatives. Boundary kinetic and scalar terms add local endpoint pieces, so a source-level interval target is

$$K_J^{\text{int}}(\lambda) = P_J^\dagger (K_J^{\text{DtN}}(\lambda) + K_J^{\text{brane}}(\lambda)) P_J \longrightarrow \begin{pmatrix} \lambda & -\sqrt{J} \\ -\sqrt{J} & \lambda + J \end{pmatrix}, \quad (99)$$

where P_J is the projection to $\mathcal{H}_J = \text{span}\{h_J, a_J\}$. This form makes the source-data demand explicit: the Dirichlet-to-Neumann kernel, brane terms, projection, inner product, and decoupling rule must be supplied by one interval model.

The current-channel entry can be isolated after the physical boundary kernel has been converted to the dimensionless DeVries normalization. Let $\mathcal{N}_T^\partial = \text{span}\{W_T^3, B_T\}$ carry the CHM product $\langle \cdot, \cdot \rangle_{\text{CHM}}$. The CHM-normalized projector package is

$$u_{\text{CHM}} \mapsto \left(\mathcal{N}_T^\partial, \langle \cdot, \cdot \rangle_{\text{CHM}}, P_{h,J}, P_{a,J}^{\gamma^\perp}, P_\gamma, K_{\gamma,J}^{\text{ref}}, \Lambda_{\text{CHM}}, \mathcal{R}_{\text{pole}} \right). \quad (100)$$

It uses the neutral trace definitions of Eqs. (46)–(49) with the source product specialized to $\langle \cdot, \cdot \rangle_{\text{CHM}}$. Define the unnormalized traces

$$\tilde{a}_\gamma = P_\gamma a_\gamma^{(0)}, \quad \tilde{z}_J = P_{a,J} \Pi_{\gamma^\perp}^{(T)} z^{(0)}. \quad (101)$$

The normalized traces are

$$a_\gamma = \frac{\tilde{a}_\gamma}{\sqrt{\langle \tilde{a}_\gamma, \tilde{a}_\gamma \rangle_{\text{CHM}}}}, \quad a_J^\perp = \frac{\tilde{z}_J}{\sqrt{\langle \tilde{z}_J, \tilde{z}_J \rangle_{\text{CHM}}}}. \quad (102)$$

The admissible neutral CHM trace package is the following proof checklist. The projected norms are positive,

$$\langle \tilde{a}_\gamma, \tilde{a}_\gamma \rangle_{\text{CHM}} > 0, \quad \langle \tilde{z}_J, \tilde{z}_J \rangle_{\text{CHM}} > 0. \quad (103)$$

The photon-orthogonal current sample remains orthogonal to the normalized photon trace,

$$\left\langle \frac{\tilde{z}_J}{\sqrt{\langle \tilde{z}_J, \tilde{z}_J \rangle_{\text{CHM}}}}, \frac{\tilde{a}_\gamma}{\sqrt{\langle \tilde{a}_\gamma, \tilde{a}_\gamma \rangle_{\text{CHM}}}} \right\rangle_{\text{CHM}} = 0. \quad (104)$$

The source theory supplies the ordered product $P_{a,J} \Pi_{\gamma^\perp}^{(T)}$, or it supplies the commutation theorem

$$[P_{a,J}, \Pi_{\gamma^\perp}^{(T)}] \big|_{\mathcal{N}_T^\partial} = 0. \quad (105)$$

In the current-entry formulas below a_J abbreviates $a_J^{\gamma^\perp}$. The transverse kernel is

$$K_{T,J} = K_{T,J}^{\text{DtN}} + K_{T,J}^{\text{brane}}.$$

Photon-reference subtraction and the source scale are

$$P_\gamma^\dagger (K_{T,J}(0) - K_{\gamma,J}^{\text{ref}}) P_\gamma = 0, \quad \hat{\lambda} = \frac{\lambda}{\Lambda_{\text{CHM}}^2}. \quad (106)$$

With endpoint-normal signs and dimensionful brane terms fixed, define

$$\hat{K}_J^{\text{cur}}(\hat{\lambda}) = \frac{1}{\Lambda_{\text{CHM}}^2} \langle a_J, (K_{T,J}^{\text{DtN}}(\lambda) + K_{T,J}^{\text{brane}}(\lambda) - K_{\gamma,J}^{\text{ref}}(\lambda)) a_J \rangle_{\text{CHM}}, \quad \lambda = \Lambda_{\text{CHM}}^2 \hat{\lambda}. \quad (107)$$

The current kinetic slope is a theorem datum,

$$Z_J^{\text{cur}} = \partial_{\hat{\lambda}} \hat{K}_J^{\text{cur}}(0) > 0. \quad (108)$$

The canonically sloped current kernel is

$$\hat{K}_{J,\text{can}}^{\text{cur}} = (Z_J^{\text{cur}})^{-1} \hat{K}_J^{\text{cur}}, \quad \partial_{\hat{\lambda}} \hat{K}_{J,\text{can}}^{\text{cur}}(0) = 1. \quad (109)$$

The constant entry in the unrescaled normalization is equivalently

$$\hat{\Sigma}_{aa,J}^{\text{CHM}} = \frac{1}{\Lambda_{\text{CHM}}^2} [\langle a_J, K_{T,J}(0) a_J \rangle_{\text{CHM}} - \langle a_\gamma, K_{T,J}(0) a_\gamma \rangle_{\text{CHM}}] a_J^{\gamma^\perp} = \frac{P_{a,J} \Pi_{\gamma^\perp}^{(T)} z^{(0)}}{\sqrt{\langle P_{a,J} \Pi_{\gamma^\perp}^{(T)} z^{(0)}, P_{a,J} \Pi_{\gamma^\perp}^{(T)} z^{(0)} \rangle_{\text{CHM}}}}, \quad a_\gamma = \frac{P_\gamma a_\gamma^{(0)}}{\sqrt{\langle P_\gamma a_\gamma^{(0)}, P_\gamma a_\gamma^{(0)} \rangle_{\text{CHM}}}}. \quad (110)$$

The source-supported CHM statement is the existence of this transverse boundary kernel from the action, gauge fixing, boundary variation, and modified product. The open DeVries-specific theorem target is

$$\hat{K}_{J,\text{can}}^{\text{cur}}(\hat{\lambda}) = \hat{\lambda} + \hat{\Sigma}_{aa,J}^{\text{CHM,can}} + O(\hat{\lambda}^2), \quad \hat{\Sigma}_{aa,2}^{\text{CHM,can}} = 2. \quad (111)$$

The target has ordered source inputs. The CHM source equations must define the neutral transverse trace space, the gauge-sector product on $\mathcal{N}_T^\partial$, and the boundary kernels $K_{T,J}^{\text{DtN}}$ and $K_{T,J}^{\text{brane}}$. Photon reference subtraction must fix P_γ and $K_{\gamma,J}^{\text{ref}}$. The current projection $P_{a,2}^{\gamma^\perp}$, the

shared source scale Λ_{CHM} , and the kinetic slope Z_2^{cur} are the data required to determine

$$\left\{ P_\gamma, P_{a,2}^{\gamma^\perp}, K_{\gamma,2}^{\text{ref}}, \Lambda_{\text{CHM}}, Z_2^{\text{cur}}, \langle \cdot, \cdot \rangle_{\text{CHM}} \right\} \longrightarrow \hat{\Sigma}_{aa,2}^{\text{CHM,can}}. \quad (112)$$

The acceptance criterion is Eq. (111). The rejection criteria are

$$\hat{\Sigma}_{aa,2}^{\text{CHM,can}} \neq 2, \quad Z_2^{\text{cur}} \leq 0, \quad \langle a_2^{\gamma^\perp}, a_2^{\gamma^\perp} \rangle_{\text{CHM}} \leq 0, \quad \langle a_2^{\gamma^\perp}, a_\gamma \rangle_{\text{CHM}} \neq 0. \quad (113)$$

These criteria classify the interval construction in the chosen source convention. A rejection outcome leaves a neighboring boundary-kernel model with a different Δ_Y^{CHM} , source scale, or projector normalization. This is the CHM current-entry theorem target. The corresponding source equations are the CHM boundary kinetic equations, modified product, vector Robin equations, scalar A_5/π_i boundary equations, and photon/custodial boundary conditions, specifically the groups recorded in Eqs. (2.11)–(2.23), (2.30)–(2.44), and (3.32)–(3.43) of Ref. [7]. Promotion of Eq. (111) requires the source definitions of $P_{a,J}^{\gamma^\perp}$, P_γ , $K_{\gamma,J}^{\text{ref}}$, Λ_{CHM} , Z_J^{cur} , endpoint signs, the CHM product, and the pole map in a single convention.

The split assigns source data and theorem data separately. CHM supplies the boundary variation, modified scalar product, vector Robin equations, scalar A_5/π_i boundary equations, and photon/custodial accounting.

The DeVries reduction supplies additional theorem data:

$$P_{h,J}, \quad P_{a,J}^{\gamma^\perp}, \quad P_\gamma, \quad K_{\gamma,J}^{\text{ref}}, \quad \Lambda_{\text{CHM}}, \quad \mathcal{R}_{\text{pole}}.$$

The interval route requires derivations of the target conditions

$$\hat{\Sigma}_{aa,2}^{\text{CHM,can}} = 2, \quad \hat{\mathcal{I}}_{h,3/4}^{\text{CHM}} = \frac{3}{4}, \quad \Delta_Y^{\text{CHM}} = 0, \quad \Delta_W^{\text{CHM}} = 0,$$

For the Target II assignment theorem this current entry has to be paired with a charged order-parameter sample in the same source space:

$$h_J = P_{h,J}(\pi_i, A_5, \delta v_i, \delta R, \alpha_H), \quad (114)$$

$$a_J^{\gamma^\perp} = \frac{P_{a,J} \Pi_{\gamma^\perp}^{(T)} z^{(0)}}{\sqrt{\langle P_{a,J} \Pi_{\gamma^\perp}^{(T)} z^{(0)}, P_{a,J} \Pi_{\gamma^\perp}^{(T)} z^{(0)} \rangle_{\text{CHM}}}}, \quad a_\gamma = \frac{P_\gamma a_\gamma^{(0)}}{\sqrt{\langle P_\gamma a_\gamma^{(0)}, P_\gamma a_\gamma^{(0)} \rangle_{\text{CHM}}}}. \quad (115)$$

Here π_i denotes boundary Goldstone data, A_5 the fifth component of the gauge field, δv_i boundary scalar-vev variations, δR radion or interval-size variations, and α_H a Wilson-line/Hosotani coordinate when the route uses one. The ordered invariant target is

$$\hat{\mathcal{I}}_{h,J_H}^{\text{CHM}} = J_H = \frac{3}{4}, \quad \hat{\Sigma}_{aa,J_{\text{adj}}}^{\text{CHM,can}} = J_{\text{adj}} = 2, \quad (116)$$

as target equations using one Λ_{CHM} , one product, and one pole map.

Datum	Projector J	Hilbert factor	Boundary datum	Normalization	Role functional	Status
h_J	$P_{h,J}$	3/4	order parameter	$A_5, \pi_i, \delta v_i, \delta \Lambda_{\text{CHM}}, \langle \cdot, \cdot \rangle_{\text{CHM}}$	$P_W, \mathcal{R}_{\text{pole}}$	open ordered sample
$W^\pm$ sample	P_W	3/4	charged-sample target	charged boundary trace coupled to h_J	same as h_J after matching	open ordered map
Z	P_Z	2	neutral current	$a_J^{\perp}$ Robin/DtN data	$M_{Z,\text{pole}}$	current entry target
γ	P_γ	0	null direction	photon zero-mode trace	reference subtraction	zero pole source-supported null

The data still to be derived are $P_{h,J}$, $P_{a,J}^{\perp}$, P_W , P_Z , $K_{\gamma,J}^{\text{ref}}$, the source scale Λ_{CHM} , endpoint-normal conventions, the photon projection, the ordered W/Z boundary map, and the map $\hat{\lambda} \mapsto m_n^2/\Lambda_{\text{CHM}}^2 \mapsto \Pi_T^{(4)}(s)$.

D. Dictionary for the interval entries

The interval construction separates the entries of Eq. (84) into physically addressable data:

Entry	Candidate datum	source	DeVries-specific obligation
$\Sigma_{hh,J}^{\text{int}} = 0$	boundary reference traction	scalar or sub-datum	fix by a brane condition, or a normalization convention
$\Sigma_{aa,J}^{\text{int}} = J$	boundary product, Robin adjoint-current boundary datum	kinetic vector term, or in the boundary equation	identify the quadratic invariant that appears in the boundary equation
$\Sigma_{ha,J}^{\text{int}} \Sigma_{ah,J}^{\text{int}} = J$	scalar-vector overlap, or brane-localized coupling	mixing, endpoint/bulk or brane-generator product	derive the invariant from a normalized overlap or generator product
$(J_H, J_{\text{adj}}) = (3/4, 2)$	Higgs/order-parameter joint/current channel	order-channel and ad-pling	derive the ordered rule from the coupled interval problem

The table is intentionally strict. A boundary mass term by itself can produce many secular equations. The DeVries claim requires the trace and determinant data in Eq. (92) and the electroweak ordering of J -samples.

The possible origins of J are correspondingly limited:

$$J \in \{C_2(R_{\text{int}}), C_2(R_{\text{bdry}}), h_{\text{CFT}}, q_{\text{KK}}^2, \langle f_{\text{bdry}}, f_{\text{bulk}} \rangle^2\}. \quad (117)$$

Here C_2 denotes a quadratic invariant, h_{CFT} a conformal weight or current-algebra datum, q_{KK} a compact charge,

and the overlap term a normalized boundary/bulk integral. A derivation names one of these entries and derives the equality to J .

E. Gauge fields, boundary breaking, and the electroweak ray

Gauge fields on an interval carry a stronger constraint than scalar toy models. Boundary conditions that break gauge symmetry have to arise from the action and preserve the consistency of the lower-dimensional gauge theory. Csaki–Hubisz–Meade use this language for electroweak symmetry breaking by boundary conditions and for Higgsless models [7]. Their framework is relevant here because it gives a controlled setting in which massive vector modes, a photon zero mode, custodial structure, and brane-localized terms can be discussed in one spectral problem.

The DeVries construction requires the boundary model to preserve the electroweak ray. The low-energy gauge-Higgs mass matrix has the form

$$\mathcal{M}_{\text{EW}}^2(v) = v^2 \mathcal{P}(g, g'), \quad (118)$$

where v is radial and $\mathcal{P}(g, g')$ is projective. A source-derived interval theorem would produce a boundary operator whose pole condition reduces to this projective datum. Thus the interval model is required to yield a coupled charged/neutral construction:

$$\left[\begin{array}{c} SU(2)_L \text{ charged sector} \\ SU(2)_L \times U(1)_Y \text{ neutral sector} \\ \text{photon zero mode} \end{array} \right] \longrightarrow \frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2} = \frac{x_+(J_H)}{x_+(J_{\text{adj}})} + \dots \quad (119)$$

This requirement places the W and Z readings inside one gauge-Higgs or boundary system. The photon must remain the unbroken mode, and the massive neutral pole must carry the adjoint/current sample. The interval theorem reaches an exact pole statement after the matching remainder $\Delta_{\text{match}}^{(\text{int})}$ has been derived.

The corresponding proof obligation is

$$S_{5\text{D}} + S_{\text{bdry}} \longrightarrow \{\mathcal{B}_W(\lambda), \mathcal{B}_Z(\lambda), \mathcal{B}_\gamma(\lambda)\} \longrightarrow \text{pole data},$$

$$M_{\gamma,\text{pole}}^2 = \Delta_\gamma^{(\text{int})}, \quad \frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2} = \frac{x_+(3/4)}{x_+(2)} + \Delta_{\text{match}}^{(\text{int})}. \quad (120)$$

The proof must include the charged and neutral boundary equations, the photon zero mode, custodial conditions, the Ward-identity target $\Delta_\gamma^{(\text{int})} = 0$, the reduced inner product, and the pole matching chain.

The interval version of the ordered assignment theorem is:

$$\text{bulk gauge fields} + \text{boundary terms} + \text{order-parameter channel} \implies K_J^{\text{int}}(\lambda) \implies (J_W, J_Z) = (3/4, 2). \quad (121)$$

This theorem is parallel to the endpoint theorem in Sec. XI. The two routes can eventually compete: endpoint labels emphasize matrix algebra, while interval boundary conditions emphasize variational origin and pole equations.

F. Scalar partner from interval data

The negative branch also has a KK address. Witten's review notes that compactification naturally produces scalar degrees of freedom, including radius moduli and possible charged scalar modes tied to symmetry breaking questions [16]. Polchinski's circle compactification likewise identifies the compact radius with a lower-dimensional scalar and shows that changing the radius is tied to current algebra at enhanced-symmetry points [3]. Hosotani's compact gauge-field mechanism adds a Wilson-line scalar route [34]. In interval language, scalar data can appear as boundary values, Wilson-line-like data, brane-localized order parameters, or moduli of the compact geometry [29, 30].

The scalar-branch target is therefore

$$u_-(J) \in \mathcal{H}_J \quad \mapsto \quad \mathcal{F}_{\text{KK}} \in \{R(x), H^\dagger H, A_5, \text{brane scalar}, \text{localized modulus}\}, \quad (122)$$

Here $u_-(J)$ is the negative-branch eigenvector of the two-channel kernel. A derivation identifies $\mathcal{F}_{\text{KK}}$, states its gauge transformation properties, and explains its scheme relation to the low-energy Higgs/order-parameter variables.

The sc/int extension of this target uses a compact boundary-superconnection holonomy,

$$\mathcal{W}_{\text{sc}} = \text{P exp} \int_I (A_5 + \Phi_{\text{odd}}) dy, \quad (123)$$

and tests whether the dimensionless scalar Hessian

$$\widehat{\mathcal{F}}_{\text{sc}}^{\text{Hol/sc}} = \Lambda_J^{-2} \left. \frac{\partial^2 V_{\text{eff}}(\mathcal{W}_{\text{sc}})}{\partial \alpha_{\text{sc}}^2} \right|_{\alpha_*} \quad (124)$$

is the scalar partner of the same projected kernel that gives the hatted current entry and the photon subtraction.

G. KK completion criterion

The KK route is fixed by the following completion chain:

compact or interval source $\rightarrow$ boundary variational problem $\xrightarrow{\text{DeVries}}$ calculation asks for a normalization and a boundary limit in which

$$K_J^{\text{int}}(\lambda) \rightarrow \lambda^2 + J\lambda - J \rightarrow \{s_{\text{pole}}^2, \mathcal{F}_{\text{KK}}\}. \quad (125)$$

The first arrow is source-backed by the KK and interval literature. The second arrow is the DeVries determinant derivation. The last arrow is the pole-placement and scalar-branch assignment. This chain gives a concrete way to compare compactification, endpoint, and G_2 routes.

X. TOY ENDPOINT-BOUNDARY OPERATOR

The previous sections list possible mechanisms. This section turns those mechanisms into a minimal toy operator. The toy model is a derivation target with named matrix entries.

Let $h_J(x)$ denote a boundary-localized order-parameter amplitude and $a_J(x)$ denote an adjoint/current amplitude. The quadratic light-sector action is

$$S_J^{(2)} = \int d^4x \begin{pmatrix} h_J^\dagger & a_J^\dagger \end{pmatrix} \begin{pmatrix} \lambda & -\kappa_J \\ -\kappa_J & \lambda + \tau_J \end{pmatrix} \begin{pmatrix} h_J \\ a_J \end{pmatrix}. \quad (126)$$

Here λ is the spectral variable after normalization. The DeVries target is

$$\tau_J = J, \quad \kappa_J^2 = J, \quad (127)$$

which gives

$$\det \begin{pmatrix} \lambda & -\sqrt{J} \\ -\sqrt{J} & \lambda + J \end{pmatrix} = \lambda^2 + J\lambda - J. \quad (128)$$

The fields h_J and a_J are placeholders with fixed jobs. The h_J channel must carry the doublet/order-parameter localized modulus, invariant when $J \neq 3/4$. The a_J channel must carry the adjoint/current invariant when $J = 2$. The off-diagonal entry κ_J is the physical mixing amplitude between the two channels. The diagonal entry τ_J is the trace datum of the adjoint/current side.

A. Boundary origin

Interval field theory supplies a mechanism for such a matrix. Boundary mass and kinetic terms modify the variational boundary conditions, and brane-localized kinetic terms can make the boundary condition depend on the four-dimensional eigenvalue [7]. A schematic bulk-plus-boundary system is

$$S_{\text{bulk}} = \int d^4x \int_0^L dy \frac{1}{2} \left[(\partial_M A_J)^2 + (\partial_M H_J)^2 \right] + S_{\text{bdry}}[A_J, H_J]. \quad (129)$$

After solving the bulk equations and restricting to the two light boundary amplitudes, the effective inverse propagator has the form

$$K_J^{\text{eff}}(\lambda) = \begin{pmatrix} \lambda + \Sigma_{hh,J}(\lambda) & \Sigma_{ha,J}(\lambda) \\ \Sigma_{ah,J}(\lambda) & \lambda + \Sigma_{aa,J}(\lambda) \end{pmatrix}. \quad (130)$$

$$\Sigma_{hh,J} = 0, \quad \Sigma_{aa,J} = J, \quad \Sigma_{ha,J} \Sigma_{ah,J} = J. \quad (131)$$

This formulation gives each matrix entry a physical address. $\Sigma_{hh,J}$ tests whether the order-parameter channel can be used as the zero reference. $\Sigma_{aa,J}$ tests whether the adjoint/current channel supplies the trace J . $\Sigma_{ha,J}$ tests whether endpoint, brane, or overlap physics supplies $\sqrt{J}$.

B. Endpoint interpretation

In an open-string or D-brane reading, a_J is an endpoint current or Chan–Paton-like gauge excitation, while h_J is a boundary order-parameter mode. Coincident branes naturally give nonabelian gauge fields from endpoint sectors [2, 3]. The toy model demands a more specific result: the endpoint algebra must reduce to the entries in Eq. (127).

The corresponding route is

$$\begin{aligned} \text{endpoint labels} &\longrightarrow \kappa_J^2 = J \\ \text{adjoint/current trace} &\longrightarrow \tau_J = J \\ \text{boundary normalization} &\longrightarrow \Sigma_{hh,J} = 0. \end{aligned} \quad (132)$$

This table defines the calculation. It also gives a compact way to compare candidate completions.

C. Electroweak matching

The toy operator must match the electroweak projective data. The desired chain is

$$\begin{aligned} \text{boundary data} &\longrightarrow \mathcal{M}_J \longrightarrow \{x_+(J), x_-(J)\} \\ &\quad \downarrow \\ &\quad \text{W/Z pole ratio and scalar} \end{aligned} \quad (133)$$

The positive branch should yield the pole-scheme target

$$\frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2} = \frac{x_+(J_H)}{x_+(J_{\text{adj}})} + \Delta_{\text{toy}}, \quad (J_H, J_{\text{adj}}) = (3/4, 2). \quad (134)$$

Here Δ_{toy} records every step missing from the toy model: field normalization, source scale, effective matching, and pole-scheme placement. The negative branch should identify the scalar or boundary partner in the same basis. The toy model therefore joins the three open calculations: ordered electroweak sampling, determinant derivation, and scalar-branch interpretation.

XI. ENDPOINT GAUGE DATA AND BRANE HIGGSING

The toy operator in Sec. X becomes physical once its entries can be assigned to recognizable string or boundary data. Open strings and D-branes give one sharp language for this assignment. The route follows the original string-theory ambition: a particle spectrum is encoded in worldsheet and endpoint data, and spacetime fields arise as low-energy modes of that spectrum.

A. Endpoint labels as a source of matrix structure

An open string ending on a stack of N branes carries endpoint labels. In modern language these labels

are Chan–Paton data. The corresponding low-energy fields sit in $N \times N$ matrices; for coincident branes, the massless excitations form a $U(N)$ gauge field and adjoint scalar fields on the brane worldvolume [2]. Polchinski’s review states the same logic in compactification language: current algebras of the compact worldsheet theory give spacetime gauge bosons, and D-brane boundary conditions give open-string gauge fields along the brane [3].

The relevant point is the matrix origin. Endpoint labels give a natural place for diagonal and off-diagonal channels:

$$|\psi; m, n\rangle, \quad m, n = 1, \dots, N, \quad (135)$$

with fields

$$A_a = (A_a)^m{}_n, \quad \Phi^I = (\Phi^I)^m{}_n. \quad (136)$$

The diagonal entries describe strings whose endpoints lie on the same brane. The off-diagonal entries describe strings stretched between distinct branes. At coincidence the off-diagonal vector modes fill out the nonabelian gauge multiplet. This is precisely the type of two-channel structure needed by Eq. (11).

The source-backed part of the argument ends here. The DeVries-specific step is sharper and remains open: the endpoint algebra must produce the two entries

$$\kappa_J^2 = J, \quad \tau_J = J, \quad (137)$$

with $J = 3/4$ and $J = 2$ entering in the ordered electroweak way. The endpoint labels supply a matrix arena; selecting Eq. (137) requires additional dynamics.

B. Worldvolume Yang–Mills and adjoint scalars

The low-energy action for coincident D-branes contains the nonabelian Yang–Mills field strength and adjoint scalars [2]. Schematically,

$$S_{\text{brane}} = \int d^{p+1}\xi \, \text{Tr} \left[-\frac{1}{4g_{\text{YM}}^2} F_{ab} F^{ab} - \frac{1}{2} D_a \Phi^I D^a \Phi^I + \frac{1}{4} [\Phi^I, \Phi^J]^2 + \dots \right] \quad (138)$$

This schematic form is enough for the entry dictionary below. Gauge fields and scalars occupy the same matrix algebra. The covariant derivative

$$D_a \Phi^I = \partial_a \Phi^I + i[A_a, \Phi^I] \quad (139)$$

couple off-diagonal vector modes to scalar expectation values. When two branes are separated, the off-diagonal gauge fields acquire a mass proportional to the separation; equivalently, the mass is the energy of a string stretched between the branes [2]. The brane construction therefore gives a geometric Higgs mechanism with a scalar modulus and charged vector modes in one matrix system.

This structure motivates the endpoint route for the DeVries problem. The positive branch is meant to describe

a vector pole-ratio clue. The negative branch asks for a scalar or order-parameter partner. A brane worldvolume action naturally contains both vector and scalar matrix entries. The target is a two-dimensional reduction normalized to

$$\mathcal{M}_J = \begin{pmatrix} 0 & \kappa_J \\ \kappa_J & -\tau_J \end{pmatrix}, \quad \kappa_J^2 = \tau_J = J. \quad (140)$$

Endpoint data therefore supply the matrix arena. The equalities $\kappa_J^2 = \tau_J = J$, the projection to $\mathcal{H}_J$, and the branch survival rule remain source-kernel theorem data.

C. Endpoint dictionary for the DeVries entries

The endpoint hypothesis can now be stated as a dictionary. Let h_J be the boundary order-parameter amplitude and a_J the endpoint-current amplitude. Let $\mathcal{H}_J = \text{span}\{h_J, a_J\}$ be the two-dimensional light sector. A candidate derivation must provide three pieces of data:

$$\text{zero reference: } \langle h_J | K_J | h_J \rangle = 0, \quad (141)$$

$$\text{trace entry: } \langle a_J | K_J | a_J \rangle = J, \quad (142)$$

$$\text{mixing entry: } \langle h_J | K_J | a_J \rangle \langle a_J | K_J | h_J \rangle = J. \quad (143)$$

Here K_J is the normalized quadratic kernel obtained after integrating out heavier worldvolume or bulk modes. Eq. (142) should be traced to a current algebra, adjoint Casimir, or boundary kinetic term. Eq. (143) should be traced to an endpoint overlap, a brane-localized coupling, or a generator normalization. Eq. (141) fixes the baseline for the order-parameter channel.

The representation data suggest the following table:

DeVries entry	Endpoint or brane origin	Required derivation
$\tau_J = J$	adjoint/current trace, boundary kinetic term, compact current algebra	show the normalized trace is the quadratic invariant
$\kappa_J^2 = J$	endpoint overlap, stretched-string or brane/order-parameter coupling	show the off-diagonal product equals the same invariant
$\Sigma_{hh,J} = 0$	order-parameter reference channel	justify the subtraction by a symmetry, boundary condition, or renormalization convention
$(J_H, J_{\text{adj}}) = (3/4, 2)$	doublet order parameter and adjoint current	derive the ordered W/Z sampling rule

This dictionary is restrictive. A completion must determine the entries, the basis, and the normalization from the source theory.

D. Scalar branch from brane separation

The brane picture also gives a natural language for the negative branch. In the D-brane Higgs mechanism, scalar expectation values describe brane positions. Off-diagonal vector fields become massive when the branes separate. Near coincidence, the off-diagonal modes become light and the full matrix structure controls the local physics [2]. Thus a two-branch matrix can plausibly pair a vector branch with scalar-modulus data, brane-separation data, auxiliary data, or effective brane-scaling data.

For the DeVries construction, this becomes the following target. There should be a scalar or boundary modulus φ_J and a normalized kernel K_J such that

$$K_J|_{\mathcal{H}_J} \sim \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}, \quad \varphi_J \text{ is read from } u_-(J). \quad (144)$$

Here $u_-(J)$ is the eigenvector of the negative eigenvalue. The phrase “read from” is an obligation: it must become a gauge-invariant map from the eigenvector data to a scalar quantity such as $H^\dagger H$, a brane separation, a boundary modulus, or a Higgs-potential parameter.

The scalar branch should also respect the electroweak ray. If the compact or brane model is meant to reproduce the Standard Model Higgs mechanism, the radial vacuum parameter v controls the common mass scale while the DeVries quotient controls the projective direction. A brane-separation interpretation must therefore identify which modulus supplies the radial scale and which normalized matrix supplies the angular data.

E. Endpoint completion criterion

The endpoint route makes the proposal more constrained. It asks for an explicit reduction:

$$\begin{aligned} \text{endpoint data} &\longrightarrow \text{worldvolume kernel} \longrightarrow \mathcal{M}_J, \\ \mathcal{M}_J &\longrightarrow \{x_+(J), x_-(J)\} \longrightarrow \text{pole ratio and scalar branch} \end{aligned} \quad (145)$$

Each arrow has a check. The first arrow is source-supported by Chan–Paton labels, D-brane gauge fields, and adjoint scalars. The second arrow is the DeVries-specific determinant derivation. The third arrow is the exact algebra of Sec. II. The last arrow is the pole-placement and negative-branch problem. Thus the endpoint route currently supplies matrix fields and a brane-Higgsing arena. Completion requires a worldvolume quadratic kernel, a projection to $\mathcal{H}_J$, an inner product, and a normalization deriving $\kappa_J^2 = \tau_J = J$ from the endpoint source theory.

This is also where the seventies lineage becomes substantive. The original dual amplitude began as a spectral object for hadron physics, and open-string amplitudes expose a tower of particle poles [2]. The technical pattern is a spectral clue, followed by the worldsheet or boundary object that produces it, followed by translation into the electroweak pole scheme. The general string-theory idea that particle data can arise from endpoint and compactification spectra is source-backed; the electroweak translation is speculative.

XII. G_2 LOCALIZATION AND LOCAL SPECTRAL OPERATORS

The G_2 route supplies a local language for the DeVries program. Its role in this paper is local. It gives a framework in which gauge fields, charged matter, order-parameter data, and deformation modes can live at singular loci of a compactification. M-theory on G_2 -holonomy spaces is a standard route to four-dimensional $\mathcal{N} = 1$ structure; singularities provide the gauge and matter sectors needed for particle physics [17–19]. The DeVries question is whether a localized G_2 operator can reduce to the same two-channel determinant used in the pole-ratio clue.

A. Singular support for gauge and matter sectors

The source-backed starting point is the localization of degrees of freedom. Witten explains that G_2 compactifications with singularities can support abelian or non-abelian gauge groups with chiral fermions, with anomaly inflow from the bulk supplying a consistency condition [17]. Acharya and Witten describe examples where singular loci carry nonabelian gauge groups and conical or intersection points support chiral multiplets [18]. Acharya and Gukov review the broader picture: singular G_2 spaces are the essential setting for nonabelian gauge groups, chiral fermions, and strongly coupled gauge dynamics in M-theory compactifications [19].

The DeVries problem asks for a local sector with three named pieces:

$$\begin{pmatrix} \text{singular gauge locus} \\ \text{localized charged matter or order parameter} \\ \text{deformation mode of the singularity} \end{pmatrix} \rightsquigarrow \mathcal{M}_{\text{localized representation data}} \quad (146)$$

The singular gauge locus is the natural address for adjoint/current data. The localized charged matter or order-parameter mode is the natural address for the doublet side. The deformation mode is the natural address for the scalar branch.

This local focus also controls the topology issue. A compact G_2 model imposes global constraints, including anomaly, representation, and line-operator data. The determinant can be tested locally first. A global compact-

ification can then be asked to support the same local operator and the Standard Model global form.

B. Higgs-bundle variables on an associative three-cycle

Modern local G_2 gauge sectors can be organized by a Higgs-bundle-like system on an associative three-cycle M_3 . Braun, Cizel, Hubner, and Schafer-Nameki describe a gauge field together with a one-form Higgs field on M_3 , with matter localized at critical loci and interactions controlled by gradient-flow data [20]. In the abelianized local description, the Higgs field can be written schematically as

$$\Phi = \sum_{i=1}^n t_i df_i, \quad \rho = \sum_{i=1}^n t_i \rho_i, \quad \Delta f_i = \rho_i, \quad (147)$$

where t_i are Cartan generators and the ρ_i encode charge distributions. For a representation with charge vector $Q = (q_1, \dots, q_n)$, the effective Morse function is

$$f_Q = \sum_{i=1}^n q_i f_i, \quad \rho_Q = \sum_{i=1}^n q_i \rho_i. \quad (148)$$

Matter in the representation R_Q localizes at points where

$$df_Q(p) = 0. \quad (149)$$

Flow lines between such critical points can generate mass terms, and trivalent flow trees can generate Yukawa couplings when the representation charges sum to zero [20]. The resulting local dictionary expresses gauge data, matter localization, mass terms, and interactions as functions of the same geometric input.

For the DeVries construction, the Higgs-bundle variables give a possible address for the local operator:

$$\mathcal{O}_J^{G_2} = \mathcal{O}_J(W, \Phi, \rho_Q, f_Q), \quad (150)$$

where W is the gauge field on M_3 . The precise form of $\mathcal{O}_J^{G_2}$ is open. The variables W , Φ , and f_Q already separate adjoint gauge data, scalar deformation data, and localized representation data.

The same Morse data suggest a Hodge/SUSY-QM refinement. Witten's construction uses $d_t = e^{-th} de^{th}$, its adjoint, and the associated deformed Laplacian to localize low modes near critical points of a Morse function [36]. In the local G_2 setting, f_Q is already the source-backed Morse function for matter localization. A candidate DeVries operator may therefore begin with a finite reduction of $(\Omega^0, \Omega^1, d_t, d_t^\dagger)$ on the singular support. The local derivation must then supply the breaking entry, the two-channel projection, and the electroweak assignment inside the same G_2 model.

C. Local two-channel kernel

Let a_J be an adjoint/current fluctuation supported along the singular gauge locus, and let h_J be a localized order-parameter fluctuation at a critical point of f_Q . Let

$$\mathcal{H}_J^{G_2} = \text{span}\{h_J, a_J\}. \quad (151)$$

After integrating out heavier localized and bulk modes, the local inverse propagator or spectral kernel has the same abstract form as the endpoint and interval kernels:

$$K_J^{G_2}(\lambda) = \begin{pmatrix} \lambda + \Sigma_{hh,J}^{G_2}(\lambda) & \Sigma_{ha,J}^{G_2}(\lambda) \\ \Sigma_{ah,J}^{G_2}(\lambda) & \lambda + \Sigma_{aa,J}^{G_2}(\lambda) \end{pmatrix}. \quad (152)$$

The DeVries target is

$$\Sigma_{hh,J}^{G_2} = 0, \quad \Sigma_{aa,J}^{G_2} = J, \quad \Sigma_{ha,J}^{G_2} \Sigma_{ah,J}^{G_2} = J. \quad (153)$$

The source-equation check for this target is explicit:

$$\Phi = \sum_i t_i df_i, \quad \rho = \sum_i t_i \rho_i, \quad \Delta f_i = \rho_i, \quad (154)$$

and

$$f_Q = \sum_i q_i f_i, \quad \rho_Q = \sum_i q_i \rho_i, \quad df_Q(p) = 0. \quad (155)$$

The current entry $\Sigma_{aa,J}^{G_2} = J$ is tested by the following local ADE-pairing protocol. Fix a source datum

$$u_{G_2}^{\text{loc}} = (M_3, \Gamma_{\text{ADE}}, W, \Phi, \rho_i, f_i, Q, p, \gamma, A_1 \subset A_2, \Lambda_{G_2}), \quad (156)$$

where the $A_1 \subset A_2$ enhancement records the local root-pairing test

$$\text{ad } A_2 \downarrow_{A_1 \times U(1)} = (\text{ad } A_1)_0 \oplus \mathbf{1}_0 \oplus \mathbf{2}_{+q} \oplus \mathbf{2}_{-q}. \quad (157)$$

The singular support carries the pairing

$$\langle \eta, \zeta \rangle_{G_2, u} = \int_{M_3} \text{Tr}_{\text{ADE}}(\eta \wedge *_{M_3} \zeta) + \sum_{p \in Z(df_Q)} \eta(p)^\dagger G_Q(p) \zeta(p), \quad (158)$$

The first term is the current-support pairing on the ADE locus. The point support term is the localized matter metric required by the later flow-overlap entry. The metric $G_Q(p)$ is theorem data: it must be positive on the localized charge sector, gauge covariant, compatible with the charge vector Q , and reducible to the wavefunction or Hessian normalization at an isolated Morse critical point. A Morse–Bott critical component replaces the point metric by the corresponding component metric and flow-moduli convention.

The local off-diagonal target is a flow-overlap theorem. Let $\psi_{h,J}^{G_2}$ denote the localized order-parameter wavefunction and let $a_J^{G_2}$ denote the projected ADE current fluctuation. Their source normalizations are

$$\langle \psi_{h,J}^{G_2}, \psi_{h,J}^{G_2} \rangle_{G_Q} = 1, \quad \langle a_J^{G_2}, a_J^{G_2} \rangle_{G_2, u} = 1. \quad (159)$$

For an oriented flow line or flow-tree contribution γ , introduce a flow kernel $\mathcal{K}_{\text{flow}}^{G_2}(\gamma)$, orientation sign ϵ_γ , and bilinear maps $B_{ha}^{G_2}$ and $B_{ah}^{G_2}$. The hatted off-diagonal entries are theorem data:

$$\begin{aligned} \hat{\Sigma}_{ha,J}^{G_2} &= \Lambda_{G_2}^{-1} \sum_\gamma \epsilon_\gamma B_{ha}^{G_2} \left(\psi_{h,J}^{G_2}, \mathcal{K}_{\text{flow}}^{G_2}(\gamma) a_J^{G_2} \right), \\ \hat{\Sigma}_{ah,J}^{G_2} &= \Lambda_{G_2}^{-1} \sum_\gamma \epsilon_\gamma B_{ah}^{G_2} \left(a_J^{G_2}, \mathcal{K}_{\text{flow}}^{G_2}(\gamma) \psi_{h,J}^{G_2} \right). \end{aligned} \quad (160)$$

The flow-overlap target is

$$\hat{\Sigma}_{ha,J}^{G_2} \hat{\Sigma}_{ah,J}^{G_2} = J. \quad (161)$$

The source datum is the existence of localized modes at $df_Q = 0$, oriented flow-line mass terms, and trivalent flow-tree interaction data. The theorem specifies how the quadratic flow-line bilinear is extracted from those data. The DeVries kernel then uses the product in Eq. (161). The source-backed two-point object in Braun–Cizel–Hubner–Schafer–Nameki is the Morse–Witten mass matrix

$$M_{\text{MW}}^{ab} = \sum_\gamma n_\gamma \exp[-tq(f(p_a) - f(p_b))], \quad n_\gamma = \pm 1, \quad (162)$$

acting between localized chiral multiplet ground states. Its Morse–Bott extension keeps the same flow-line form for point-to-component trajectories and gives a zero matrix element for two states localized on the same critical circle. The current source package therefore supplies a narrower test,

$$M_{\text{MW}}^{ab} \rightsquigarrow (M_{ha}^{\text{MW}}, M_{ah}^{\text{MW}}) \quad \text{inside} \quad \mathcal{H}_J^{G_2} = \text{span}\{h_J^{G_2}, a_J^{G_2}\}. \quad (163)$$

The source test isolates the failed derivation step: the Braun source data lack an identification of the projected ADE current $a_J^{G_2}$ as a Morse–Witten endpoint state in this two-channel block. Target IIIh therefore remains active only after an additional same-basis projection theorem supplies $M_{ha}^{\text{MW}}, M_{ah}^{\text{MW}}, \Lambda_{G_2}$, and the orientation convention.

The local current test chooses a projected adjoint fluctuation $a_J^{G_2}$ with

$$\langle a_J^{G_2}, a_J^{G_2} \rangle_{G_2, u} = 1 \quad (164)$$

and defines

$$\begin{aligned} \hat{K}_J^{G_2, \text{cur}}(\hat{\lambda}) &= \Lambda_{G_2}^{-2} \left\langle a_J^{G_2}, \mathcal{K}_{\text{cur}}^{\text{ADE}}(\lambda) a_J^{G_2} \right\rangle_{G_2, u}, \\ \lambda &= \Lambda_{G_2}^2 \hat{\lambda}. \end{aligned} \quad (165)$$

Here $\mathcal{K}_{\text{cur}}^{\text{ADE}}$ denotes the projected quadratic ADE-current operator induced on the singular gauge locus, including source scale, projection, and possible singular-support boundary terms. Its detailed construction from the local G_2 gauge theory is part of the theorem data. The canonical current kernel is fixed by

$$Z_J^{G_2} = \partial_{\hat{\lambda}} \hat{K}_J^{G_2, \text{cur}}(0) > 0, \quad \hat{K}_J^{G_2, \text{cur}, \text{can}} = (Z_J^{G_2})^{-1} \hat{K}_J^{G_2, \text{cur}}. \quad (166)$$

The target is

$$\widehat{K}_J^{G_2, \text{cur}, \text{can}}(\widehat{\lambda}) = \widehat{\lambda} + \widehat{\Sigma}_{aa, J}^{G_2, \text{can}} + O(\widehat{\lambda}^2), \quad \widehat{\Sigma}_{aa, 2}^{G_2, \text{can}} = 2. \quad (167)$$

This local test precedes the off-diagonal flow-overlap product and the electroweak pole map. Compact global completion, anomaly inflow, the charge lattice, hypercharge embedding, and pole matching remain theorem data. Appendix C 3 states the same protocol as Target IIIg.

D. Dictionary for the G_2 entries

The local G_2 dictionary is:

DeVries try	en- G_2 origin	Derivation target
$\Sigma_{aa, J}^{G_2} = J$	ADE $A_1 \subset A_2$ hancement singularity- pairing	singular derive the canon- gauge sector with ical current entry en- $\widehat{\Sigma}_{aa, 2}^{G_2, \text{can}} = 2$ and singular-support
$\Sigma_{ha, J}^{G_2} \Sigma_{ah, J}^{G_2} = J$	localized ter at $df_Q = 0$, flow-line coupling, bifundamental vev, support or singularity- deformation mixing functions	mat- derive the flow- product overlap from the point- metric and local wave- functions
$\Sigma_{hh, J}^{G_2} = 0$	reference parameter channel entry through lo- cal point cal normalization of f_Q	justify the zero entry through lo- cal normalization or symmetry of the critical-point problem
$(J_H, J_{\text{adj}}) = (3/4, 2)$	localized doublet/order- parameter data, adjoint/current data, and photon- orthogonal comple- tion data	derive P_W , P_Z , P_γ , hypercharge, charge lattice, and pole map in the same compact model

This table turns G_2 geometry into a determinant problem. Each geometric object must provide one entry of $K_J^{G_2}$, one assignment rule, or one consistency check. A G_2 route that supports the pole-ratio claim has to derive at least one of $\Sigma_{aa, J}^{G_2}$, $\Sigma_{ha, J}^{G_2} \Sigma_{ah, J}^{G_2}$, or $\Sigma_{hh, J}^{G_2}$ from Higgs-bundle or localized-singularity data with a stated inner product. A G_2 route enters the dimensional interpolation after the local operator is embedded into a compact model whose chiral spectrum, anomaly inflow, global form, and electroweak charge normalization are stated.

The normalization target includes the local inner product on the two-channel space,

$$\mathcal{H}_J^{G_2} = \text{span}\{h_J^{G_2}, a_J^{G_2}\}, \quad \langle h_J^{G_2}, a_J^{G_2} \rangle_{G_2}, \quad (168)$$

where the pairing is the singular-support form of Eq. (158). The current-entry derivation identifies the gauge-field fluctuation W along the singular gauge locus and then evaluates

$$W|_{M_3} \rightsquigarrow \widehat{K}_J^{G_2, \text{cur}, \text{can}} \rightsquigarrow \widehat{\Sigma}_{aa, 2}^{G_2, \text{can}} = 2. \quad (169)$$

The rejection outputs are

$$\widehat{\Sigma}_{aa, 2}^{G_2, \text{can}} \neq 2, \quad Z_2^{G_2} \leq 0, \quad \langle P_{a, 2}^{G_2} a_2, P_{a, 2}^{G_2} a_2 \rangle_{G_2, u} \leq 0, \quad (170)$$

or an unresolved extra current channel. These outputs classify the local ADE pairing before a compactification or pole claim is made. The off-diagonal-product derivation identifies the localized matter and Higgs-bundle data,

$$(\Phi, \rho_Q, f_Q, \psi_{h, J}^{G_2}, \mathcal{K}_{\text{flow}}^{G_2}, \epsilon_\gamma) \rightsquigarrow \widehat{\Sigma}_{ha, J}^{G_2} \widehat{\Sigma}_{ah, J}^{G_2} = J, \quad (171)$$

through the bilinear definitions of Eq. (160). Trivalent flow trees supply interaction terms; the quadratic kernel requires the two-point flow-line bilinear and its same-basis normalization. The rejection outputs are

$$\widehat{\Sigma}_{ha, J}^{G_2} \widehat{\Sigma}_{ah, J}^{G_2} \neq J, \quad G_Q \leq 0,$$

orientation sum lacks a source convention, extra coupled light c
(172)

The order-parameter reference is derived from the critical-point normalization of f_Q ,

$$df_Q(p) = 0 \rightsquigarrow \Sigma_{hh, J}^{G_2} = 0. \quad (173)$$

This is the local G_2 analogue of the interval Schur-complement target: the source theory must provide the three named entries before the singularity language has derivational force.

E. Higgsing as singular deformation

Acharya and Witten give a pattern directly relevant here: in examples where singular loci meet, gauge groups live on the singular components, chiral multiplets can be supported at the meeting point, and Higgsing corresponds to a geometric deformation or smoothing of the singularity [18]. In their brane-related examples, a vev for a bifundamental field can break a product gauge symmetry to a diagonal subgroup, while the local singular geometry changes accordingly.

For DeVries, this suggests the deformation chain

$$\begin{aligned} &\text{intersecting singular loci} \longrightarrow \text{localized chiral/order-parameter mode} \\ &\longrightarrow \text{deformed singularity} \longrightarrow K_J^{G_2}(\lambda). \end{aligned} \quad (174)$$

The scalar branch would then be read from the deformation direction. If $u_-(J)$ is the negative-branch eigenvector

tor in $\mathcal{H}_J^{G_2}$, the target scalar map is

$$u_-(J) \mapsto \mathcal{F}_{G_2},$$

$$\mathcal{F}_{G_2} \in \left\{ \begin{array}{l} \text{deformation parameter, } f_Q(p), \rho_Q, \\ \text{bifundamental vev, localized modulus} \end{array} \right\}. \quad (175)$$

The map must also respect the electroweak ray. The deformation parameter can supply the scalar side, while the pole-ratio branch supplies the projective vector side.

F. Anomaly and global checks

The G_2 route has consistency tests. Witten's anomaly-inflow discussion links local singularity data to four-dimensional chiral matter and anomaly cancellation [17]. Therefore a local DeVries operator has to pass the following checks:

1. the singular gauge locus supports the required adjoint sector;
2. localized charged matter has the desired representation content;
3. anomaly inflow or its local analogue is compatible with the charged fields;
4. the global gauge-group form of Sec. XV is compatible with the local representations;
5. flavor labels remain separate from compactification topology until a representation map is derived.

These checks can reject a candidate completion even when the two-channel determinant is obtained locally.

G. G_2 completion criterion

The G_2 route has the same completion chain as the endpoint and interval routes:

$$\text{singular } G_2 \text{ neighborhood} \longrightarrow (W, \Phi, \rho_Q, f_Q) \longrightarrow K_J^{G_2}(\lambda),$$

$$K_J^{G_2}(\lambda) \longrightarrow \lambda^2 + J\lambda - J \longrightarrow \{s_{\text{pole}}^2, \mathcal{F}_{G_2}, \mathcal{G}_{\text{top}}\}. \quad (176)$$

The first arrow is source-backed by singular G_2 compactification and Higgs-bundle literature. The second arrow is the DeVries determinant derivation. The final arrow is the pole-placement, scalar-branch, and global-form assignment. The local determinant is the first compatibility condition; compact global data then supply anomaly, representation, and line-operator tests.

XIII. COMPARISON OF THE THREE LOCAL ROUTES

The endpoint, interval, and G_2 sections introduce three ways to place the DeVries determinant inside a theory

with localized gauge and order-parameter data. The section casts the three routes in a common reduction format. Each route is tested by the same normalized two-channel kernel, electroweak assignment, branch interpretation, and global compatibility conditions. A derivation in any route must supply a physical origin for the matrix entries and turn the string, extra-dimensional, or singular-geometric address into a calculation.

A. Common reduction problem

Let $r \in \{\text{end}, \text{int}, G_2\}$ denote the endpoint, interval, or localized-singularity route. The corresponding source descriptions are $\mathcal{S}_{\text{end}}$, $\mathcal{S}_{\text{int}}$, and $\mathcal{S}_{G_2}$. The desired reduction chooses a two-dimensional light subspace

$$\mathcal{H}_{J,r} = \text{span}\{h_{J,r}, a_{J,r}\}, \quad (177)$$

where $h_{J,r}$ is the order-parameter channel and $a_{J,r}$ is the adjoint/current channel. In this basis the reduced spectral kernel has the form

$$K_{J,r}(\lambda) = \begin{pmatrix} \lambda + \Sigma_{hh,J}^r(\lambda) & \Sigma_{ha,J}^r(\lambda) \\ \Sigma_{ah,J}^r(\lambda) & \lambda + \Sigma_{aa,J}^r(\lambda) \end{pmatrix}. \quad (178)$$

The DeVries target is the simultaneous limit and normalization

$$\Sigma_{hh,J}^r = 0, \quad \Sigma_{aa,J}^r = J, \quad \Sigma_{ha,J}^r \Sigma_{ah,J}^r = J. \quad (179)$$

The available source equations select a candidate entry sequence. The interval/CHM route begins with the admissible neutral trace package of Eqs. (103)–(108). That package supplies positive projected norms, photon-orthogonal normalization, source-ordered or commuting projectors, photon-reference subtraction, a shared Λ_{CHM} , and the current-slope datum Z_J^{cur} . Its first theorem target is

$$\widehat{\Sigma}_{aa,2}^{\text{CHM,can}} = 2. \quad (180)$$

Sec. IX states the corresponding acceptance criteria in terms of the CHM source equations and projector data. The same source package then carries

$$\Delta_W^{\text{CHM}}, \quad \Delta_Y^{\text{CHM}}, \quad \Sigma_{ha,J} \Sigma_{ah,J} = J, \quad \Delta_{\text{O1/O10}}. \quad (181)$$

The Hodge/SUSY-QM route supplies the companion square-root theorem target. The G_2 route provides Target IIIg, a local ADE-pairing current test whose first acceptance equation is $\widehat{\Sigma}_{aa,2}^{G_2,\text{can}} = 2$, and Target IIIh, a local flow-overlap product target for $\widehat{\Sigma}_{ha,J}^{G_2} \widehat{\Sigma}_{ah,J}^{G_2} = J$. Its electroweak projectors, charge lattice, and pole map remain open obligations. The endpoint/Chan–Paton route supplies a matrix arena whose DeVries use requires a world-volume quadratic kernel, projection, and normalization. Equation (179) is stronger than a qualitative mechanism.

It assigns one geometric, endpoint, or boundary object to each matrix entry. It also fixes the algebraic route to

$$\det K_{J,r}(\lambda) = \lambda^2 + J\lambda - J, \quad \lambda \in \{x_+(J), x_-(J)\}. \quad (182)$$

The route is fully specified when one source package supplies the reduced space, source product, entry kernels, scalar functional, and pole map, and derives the route-indexed arrows in

$$\begin{aligned} (u_r, \mathcal{H}_{J,r}) &\longrightarrow (\langle \cdot, \cdot \rangle_r, \Lambda_r, P_{h,J}^r, P_{a,J}^r, K_{\text{cur}}^r, K_{\text{mix}}^r) \\ &\downarrow \\ (\mathcal{F}_{\text{sc}}^r, \mathcal{R}_{\text{pole}}^r) &\longrightarrow (\hat{\Sigma}_{aa,J}^r, \hat{\Sigma}_{ha,J}^r, \hat{\Sigma}_{ah,J}^r, P_W^r, P_Z^r, P_\gamma^r, \Delta_{\text{match}}^{(r)}). \end{aligned} \quad (183)$$

The current proof obligations include the canonical CHM neutral-current entry $\hat{\Sigma}_{aa,2}^{\text{CHM,can}} = 2$, the G_2 ADE current entry $\hat{\Sigma}_{aa,2}^{G_2,\text{can}} = 2$, the same-basis Hodge/SUSY-QM or G_2 flow-overlap product $\hat{\Sigma}_{ha,J}^r \hat{\Sigma}_{ah,J}^r = J$, and the ordered projector package P_W^r, P_Z^r, P_γ^r with its pole remainder.

The reduction has three candidate arenas. The endpoint arena is supported by the open-string fact that endpoint labels and coincident branes generate matrix-valued gauge fields, with brane positions supplying scalar data [2, 3]. The interval arena is supported by extra-dimensional models in which boundary conditions and brane-localized terms set vector spectra [7]. The G_2 arena is supported by singular compactifications that localize gauge sectors and chiral matter, together with Higgs-bundle variables organizing local matter and interaction data [17–20]. These sources support the listed ingredients. Selection of the two-channel subspace, inner product, normalization, and entries in Eq. (179) remains the derivation target.

The interval arena supplies source-derived boundary data. The CHM source equations fix the boundary kinetic normalization, the corresponding modified product, vector Robin data from boundary scalar vevs, scalar A_5/π_i boundary equations, and photon/custodial accounting. Sec. IX states the convention data. In compact route notation,

$$\begin{aligned} \partial_y f_\lambda &= -(\lambda/M_i) f_\lambda, \\ (f, g)_{\text{CHM}} &= \int fg + \sum_i M_i^{-1} f(y_i) g(y_i), \\ \partial_y A_\mu &\mp v_i^2 A_\mu = 0. \end{aligned} \quad (184)$$

Thus the source-backed statement is

$$\begin{aligned} \text{CHM boundary data} &\implies K_{\text{int}}(\lambda) \text{ exists,} \\ K_{\text{int}}(\lambda) &\implies \begin{pmatrix} \lambda & -\sqrt{J} \\ -\sqrt{J} & \lambda + J \end{pmatrix} \text{ is open.} \end{aligned} \quad (185)$$

Equation (184) supplies boundary ingredients; the projection to $\mathcal{H}_J$ and the J -normalization are theorem data. The interval route has explicit vector and scalar boundary equations for an entry-level source target. The entry

tests have two linked forms: Hodge/SUSY-QM factorization tests $\Sigma_{ha,J} \Sigma_{ah,J} = J$, and the canonical CHM boundary-current test $\hat{\Sigma}_{aa,2}^{\text{CHM,can}} = 2$. Sec. IX C defines the neutral trace package, photon subtraction, Z_J^{cur} , canonical current kernel, and target equation. The CHM equations supply the boundary variation, product, vector Robin data, scalar A_5/π_i equations, custodial accounting, and photon-zero accounting. The projection $P_{a,J}$, reference subtraction, source scale Λ_{CHM} , endpoint-normal conventions, ordered W/Z boundary map, and pole map remain theorem data. The parent KK-fixing notes sharpen the same issue into a single-source test: the dimensional middle line, negative-branch scalar reading, and electromagnetic endpoint can be combined when one source variable supplies

$$u \mapsto (t_{\text{dim}}(u), t_{\text{EW}}(u), K_J(u, \lambda), \mathcal{F}_{\text{sc}}(u; J)). \quad (186)$$

The scalar and electromagnetic readings are downstream of the current entry, ordered W/Z assignment, source-to-pole calculation, and photon-zero normalization. Target C 4 records the O17 endpoint coupling package after those data are supplied.

The common reduction is filtered by three admissibility tests. First, trace and coefficient statements stay inside the matrix where they are derived until a common source operator and field normalization identify the trace spaces. Second, Wigner–Eckart or Clebsch routes must specify the parent representation, operator tensor type, projection, and normalization. The tested doublet-parent route is removed by the ordinary selection rule. The triplet axial route gives the wrong parent-Casimir datum for Eq. (179). Third, an $\text{SO}(32)$ -flavor completion must show how its representation sector couples to the electroweak kernel, with extra-state projection and anomaly constraints stated before any flavor claim is used in the mass-ratio argument.

For the superconnection/CHM route, these filters have concrete meanings. The odd Higgs trace space, the even transverse-current trace space, and the interval boundary scalar product stay separated until one source operator defines their common inner product. The Wigner–Eckart route is a failed subroute, so the superconnection route must supply its own projection. The $\text{SO}(32)$ material remains a flavor-boundary sector until a derived coupling inserts that representation sector into the electroweak kernel.

B. Entry-by-entry comparison

The first entry is the diagonal current contribution,

$$a_{J,r} \rightsquigarrow \Sigma_{aa,J}^r = J. \quad (187)$$

The candidate source objects are a Chan–Paton or brane gauge amplitude, a boundary value of a bulk gauge mode, or a fluctuation along a singular gauge locus.

The second entry is the off-diagonal product,

$$h_{J,r} \leftrightarrow a_{J,r}, \quad \Sigma_{ha,J}^r \Sigma_{ah,J}^r = J. \quad (188)$$

The candidate source objects are a stretched open string or endpoint overlap, a boundary overlap between an order parameter and a bulk gauge profile, or a localized matter/deformation coupling.

The third entry is the order-parameter reference contribution,

$$h_{J,r} \rightsquigarrow \Sigma_{hh,J}^r = 0. \quad (189)$$

The candidate source objects are a boundary reference channel, a normalized boundary order-parameter self-energy, or a critical-point normalization of a localized mode. Appendix C 3 records the entry-by-entry proof obligations.

The following table records the three entries in a common dictionary.

Kernel entry	Endpoint/D-brane route	Interval route	KK G_2 route
$\Sigma_{aa} = J$	Chan-Paton current or brane gauge amplitude	CHM or any kernel get source conventions; J -normalization open boundary scalar A_5/π_i equa- tion, and oriented flow modified kernel, bilin-product; open ear maps, and product normalization metric critical-point normalization term; with local-to-open subtraction rule	Target $A_1 \subset A_2$ ADE current with pairing with $\widehat{\Sigma}_{aa,2}^{G_2,\text{can}} = 2$
$\Sigma_{ha}\Sigma_{ah} = J$	stretched string or endpoint overlap	boundary scalar A_5/π_i equa- tion, and oriented flow modified kernel, bilin-product; open ear maps, and product normalization metric critical-point normalization term; with local-to-open subtraction rule	Target IIIh: vev, localized matter at $df_Q = 0$, and oriented flow modified kernel, bilin-product; open ear maps, and product normalization metric critical-point normalization term; with local-to-open subtraction rule
$\Sigma_{hh} = 0$	order-parameter reference channel	boundary scalar reference open subtraction rule	critical-point normalization term; with local-to-open subtraction rule

C. Electroweak assignment as a route-independent target

The same electroweak assignment target applies to every route. The low-energy field-theory data contain an $SU(2)_L$ Higgs doublet and adjoint gauge currents. Their available quadratic invariants give

$$J_H = \frac{3}{4}, \quad J_{\text{adj}} = 2. \quad (190)$$

The target map is

$$\mathcal{A}_r : (\mathcal{S}_r, \mathcal{H}_{J,r}) \longrightarrow (J_W, J_Z) = (J_H, J_{\text{adj}}). \quad (191)$$

The ordered assignment is a theorem target. The observed W and Z vector bosons are gauge-sector particles after symmetry breaking. The quotient assigns the charged comparison to the order-parameter invariant and the neutral comparison to the adjoint/current invariant. The required derivation must produce this ordered pair from the gauge-Higgs coupling structure, endpoint boundary labels attached to the Higgsing field, interval boundary data that distinguish charged and neutral sectors, or a localized singularity model where doublet and adjoint data meet. The Standard Model mass matrix supplies the projective ratio and the representation data; the ordered sampling maps remain additional theorem data.

If the ordered assignment and pole-placement theorem hold, the pole-ratio reading has one route-labeled form:

$$\frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2} = \frac{x_+(J_H)}{x_+(J_{\text{adj}})} + \Delta_{\text{match}}^{(r)}. \quad (192)$$

Here r labels the endpoint, interval, or G_2 route. The exact pole reading requires a derived $\Delta_{\text{match}}^{(r)} = 0$ in the chosen pole scheme, or a computed remainder whose size and sign are fixed by the source-to-EFT-to-pole chain. A compactification-scale or unification-scale derivation makes the quotient a boundary datum; the pole claim then requires the matching chain.

D. Branch interpretation and scalar data

Every route diagonalizes the same two-channel kernel. Therefore each route has to account for both eigenvectors. Let

$$u_{\pm,J,r} = \alpha_{\pm,J,r} h_{J,r} + \beta_{\pm,J,r} a_{J,r} \quad (193)$$

be normalized eigenvectors of the route kernel in the De-Vries limit. The positive branch supplies the pole-ratio clue. The negative branch supplies the scalar-side question,

$$u_{-,J,r} \rightsquigarrow \mathcal{F}_{\text{sc},r}. \quad (194)$$

The functional $\mathcal{F}_{\text{sc},r}$ depends on the route. Endpoint, interval, and G_2 sources provide candidate scalar addresses: brane or boundary moduli, localized scalar modes, and singularity-deformation data. These addresses acquire physical content after a gauge-invariant functional, normalization, and electroweak scheme are specified. Appendix C 4 states the same-source condition: the first five entries of the scalar package must agree with the vector pole package, including $\mathcal{H}_J$, the inner product, the projection, Λ_J , and $\mathcal{R}_{\text{pole}}$.

The scalar map has to be tied to the electroweak ray. The gauge-Higgs system has a radial vacuum scale and a projective vector-mass ratio. The two-root structure is physically restrictive because a determinant used for the pole clue also supplies the partner branch. A source route

must identify how the positive branch fixes the projective vector datum and how the negative branch enters the radial or scalar sector.

E. Scale placement

The route comparison separates three scale placements:

$$\begin{aligned} & \text{pole scale,} \\ & \text{compactification or boundary scale,} \\ & \text{unification scale.} \end{aligned} \quad (195)$$

The construction attaches the electroweak clue to the pole scale. The attachment gives the clue its direct electroweak content. The GUT lineage supplies the comparison class: a clean weak-angle value can be a high-scale structural datum, then acquire low-energy meaning through running and thresholds. In the present construction the clean value is attached to the pole ratio, so the scale-placement theorem is logically prior to any compactification geometry.

The common matching theorem has the form

$$\mathcal{S}_r \longrightarrow K_{J,r}^{\text{bare}}(\lambda) \longrightarrow \Gamma_{\text{eff}}^{(4)}[H, W, B] \longrightarrow \Delta_{V,T}^{-1}(s)|_{s=s_{W,Z}}. \quad (196)$$

Equation (196) is the pole-placement theorem target. The first arrow is the route reduction. The second arrow matches the source kernel into a four-dimensional gauge-Higgs effective action. The third arrow places the result in the dressed transverse pole convention of Sec. III. Endpoint, interval, and G_2 routes differ in the first arrow; they share the last two arrows when the pole reading is retained. Appendix C 1 gives the equivalent transverse self-energy form of the same theorem.

For endpoint and interval routes, the central obligation is the relation between a boundary determinant and a four-dimensional pole. Boundary conditions define a spectrum before loop corrections and before conversion to pole observables. The derivation specifies how the determinant survives matching to the pole definition, or how the pole reading emerges from a dressed boundary self-energy.

For the G_2 route, a local singularity model lives at a compactification or high scale. The local kernel can organize representations and couplings. A pole-spectrum reading requires a low-energy matching chain. The resulting target is a two-stage theorem:

$$\mathcal{S}_{G_2} \longrightarrow K_{J,G_2} \longrightarrow \Delta_{V,T}^{-1}(s) \quad \text{with} \quad s = s_{W,Z}. \quad (197)$$

The first arrow is local geometry. The second arrow is four-dimensional matching. Both arrows remain open obligations.

F. Route outcome

A route succeeds when it supplies the common kernel target, the ordered electroweak assignment, the pole-matching chain, the scalar-branch map, and the global compatibility checks from one stated source theory. These criteria turn endpoint, interval, and G_2 language into comparable derivation problems. Extra light channels, a different trace, a different mixing product, a high-scale placement, or an incompatible global form would classify the DeVries relation inside a larger family of spectral constructions.

XIV. FLAVOR BOUNDARY AND REPRESENTATION ALGEBRA

The electroweak mass-ratio problem and the flavor problem carry separate obligations in this paper. The boundary condition is:

$$\begin{aligned} & \text{DeVries electroweak determinant} \\ & \text{and generation/flavor organization} \\ & \text{are linked only by an explicit model.} \end{aligned} \quad (198)$$

This condition requires any compactification or endpoint picture to give an explicit equation linking flavor data to the electroweak determinant.

Representation theory remains essential. In the Standard Model and in the familiar grand-unification lineage, fermions live in representations of the internal gauge group, while gauge bosons live in the adjoint sector [23, 24]. $SU(5)$, $Spin(10)$, Pati–Salam, and E_6 show how a larger algebra can unify the Standard Model representation data. This is the correct role of the GUT material in the present paper: it supplies representation discipline and a comparison class for structural weak-angle statements.

The DeVries construction uses a different representation question. The electroweak determinant asks why one sample is associated with the Higgs/order-parameter doublet invariant and the other with the adjoint/current invariant:

$$J_H = T_H(T_H + 1) = 3/4, \quad J_{\text{adj}} = T_{\text{adj}}(T_{\text{adj}} + 1) = 2. \quad (199)$$

Flavor representations can constrain a completion. The missing step is a derived map from endpoint, compactification, or localized matter data into the two-channel electroweak operator.

The $SO(32)$ -flavor idea is therefore an organizing boundary sector in this paper. In an endpoint language it may label external or internal flavor degrees of freedom, Chan–Paton-like data, or a representation algebra for fermion multiplicities. The representation data contain the proposed branching data

$$16 = (5, 3) + (1, 1), \quad \wedge^2(5, 3) = (15, \bar{3}) + (\bar{10}, 6). \quad (200)$$

These equations currently function as representation constraints. Their use requires a primary source or a derived embedding in which the endpoint or orientifold construction supplies the gauge group, projection, massless spectrum, and coupling to the electroweak operator. Its required checks are concrete:

1. identify the representation space carrying the $SO(32)$ -flavor labels;
2. map the Standard Model chiral fermions into that space;
3. specify orientifold, tadpole, or endpoint constraints if a Type-I-like $SO(32)$ setting is used;
4. project, lift, or decouple extra states from Eq. (200);
5. show anomaly cancellation, hypercharge normalization, and global-form compatibility;
6. couple the flavor sector to the electroweak determinant through a derived operator.

This keeps the flavor idea as a compatibility sector. The electroweak ratio concerns W/Z pole structure and the Higgs/order-parameter channel. Generation structure concerns chiral matter, Yukawa textures, and possible endpoint or compactification multiplicities. A string/Kaluza–Klein completion may eventually bind them together through an equation or a local geometric construction.

The compactification version has an additional caveat. Geometry can generate gauge symmetries, chiral localization, and multiple wavefunctions with the same gauge quantum numbers [16, 18, 20]. That makes topology relevant to flavor. This paper treats generation counting as a separate task because the DeVries determinant is a two-channel electroweak object. The decisive test is whether the same local model that yields generations also yields

$$\det(\lambda - \mathcal{O}_J) = \lambda^2 + J\lambda - J \quad (201)$$

with the ordered electroweak samples in Eq. (199).

The practical rule for later sections is therefore simple. Flavor material may enter as a constraint on endpoints, compactification, anomaly cancellation, and global form. It may motivate a local operator. The W/Z mass-ratio argument rests on the electroweak determinant and its pole placement.

XV. GLOBAL FORM OF THE STANDARD MODEL GAUGE GROUP

The local Standard Model gauge algebra admits several global forms. The gauge group may be written as

$$G_{\text{SM}} = \frac{SU(3)_C \times SU(2)_W \times U(1)_Y}{\Gamma}, \quad \Gamma \subset \mathbb{Z}_6, \quad (202)$$

with $\Gamma = 1, \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_6$ as the standard possibilities [21, 22]. Local perturbative fields see the Lie algebra. Line operators, magnetic sectors, theta-angle periodicities, and topological probes see the quotient.

This is relevant to the DeVries program for two reasons. First, any compactification, endpoint, or G_2 completion has to specify the global group it realizes. Second, a pole-spectrum determinant for the electroweak sector has to coexist with the same line-operator spectrum as the low-energy Standard Model. Thus a proposed $\mathcal{O}_J$ comes with a global-form compatibility condition:

$$\mathcal{O}_J \text{ is defined in a theory with } G_{\text{SM}} = (SU(3) \times SU(2) \times U(1))/\Gamma. \quad (203)$$

Tong’s line-operator analysis classifies how the quotient changes the electric, magnetic, and dyonic line sectors [21]. The quotient restricts admissible electric line operators and changes the allowed magnetic line operators through Dirac quantization. It also changes theta-angle periodicities. After electroweak symmetry breaking, the line sectors descend to electromagnetic probes, so Γ leaves a low-energy topological signature.

Hsin and collaborators propose a transport-style probe of this global form through a fractional topological response coefficient [22]. The important point here is structural: the global form is, in principle, observable through nonlocal or topological response data. It supplies a completion constraint for the DeVries program, because a string or compactification model generally determines more than a local Lie algebra.

Grand unification provides the comparison class. $SU(5)$ and $\text{Spin}(10)$ naturally package Standard Model representations in a way that favors the $\mathbb{Z}_6$ quotient, and the subgroup $S(U(2) \times U(3))$ realizes the quotient structure inside $SU(5)$ [23, 24]. This places the GUT lineage beside the DeVries pole reading: high-scale structural placement is already known in unified models, while the DeVries construction assigns its clean value to the low-energy pole ratio and therefore needs a separate pole-placement derivation.

The global-form constraint can be expressed as a commutative diagram requirement. If a compactification or endpoint model gives an ultraviolet group G_{UV} , then the low-energy electroweak sector should fit into

$$\begin{array}{ccc} G_{\text{UV}} & \longrightarrow & G_{\text{SM}} \\ \downarrow & & \downarrow \\ \mathcal{O}_{\text{UV}} & \longrightarrow & \mathcal{O}_J, \end{array} \quad (204)$$

where the upper arrow fixes Γ and the lower arrow reduces the ultraviolet spectral operator to the DeVries two-channel operator. A model that supplies $\mathcal{O}_J$ while leaving Γ unspecified has an underdetermined topological sector.

The mass-ratio argument itself should use local electroweak fields and pole positions. The global-form material acts as a filter on completions. A viable string/Kaluza–Klein/ G_2 derivation should state:

1. which quotient Γ appears;
2. which line operators survive compactification and electroweak breaking;
3. how the Higgs/order-parameter field transforms under the chosen global group;
4. how the DeVries operator respects the same quotient.

XVI. CONCLUSION AND OUTLOOK

The construction begins with an exact two-branch algebra. The first assumption A1 fixes

$$Q(J) = \mu^2 \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}, \quad J = s(s+1),$$

whose dimensionless eigenvalues obey $x^2 + Jx - J = 0$. The quotient

$$1 - \frac{x_+(3/4)}{x_+(2)}$$

is therefore a rigid consequence of A1 once the ordered samples $(3/4, 2)$ are chosen. The physical interpretation additionally requires A2 and A3.

The placement hypothesis A2 uses the complex-pole mass ratio

$$s_{\text{pole}}^2 = 1 - \frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2}.$$

This choice gives the spectral language a low-energy meaning and separates the proposal from $\overline{\text{MS}}$ running angles, effective weak angles, Breit-Wigner line-shape masses, and high-scale GUT boundary values. The pole placement is an assumption awaiting a dynamical pole condition. Appendix C1 states the corresponding theorem target.

The Standard Model gauge-Higgs sector supplies the projective datum

$$\frac{M_W^2}{M_Z^2} = \frac{g^2}{g^2 + g'^2}$$

and the broken-to-unbroken vacuum ray. It also supplies two representation data relevant to the proposed assignment: the Higgs/order-parameter doublet with $J_H = 3/4$, and the adjoint/current channel with $J_{\text{adj}} = 2$. The assignment hypothesis A3 is the ordered map

$$(J_W, J_Z) = (J_H, J_{\text{adj}}) = (3/4, 2)$$

from gauge-Higgs, boundary, endpoint, or compactification data. Target II is a derivation of this map from the same source data that define the two-channel operator.

The electroweak ray adds a separate admissibility condition. A source route must preserve one radial gauge-Higgs parameter, the photon null direction, and the simultaneous collapse of the charged and neutral massive vector poles at fixed projective direction. Appendix C2 records this rank-and-ray condition. Coupling-space paths and custodial-breaking shifts are separate deformation problems.

The pole convention, electroweak ray, theorem-target structure, source protocol, flavor boundary, and global-form filter serve as fixed acceptance conditions. They define the scheme, deformation class, source discipline, and completion tests that every route derivation has to satisfy. The DeVries determinant remains the dynamical target.

The negative branch belongs to the same diagonalization as the positive branch. Its scalar interpretation remains conditional on a gauge-invariant functional, a normalization, and a scheme. Candidate targets include a Higgs-potential parameter, a scalar pole, a boundary modulus, a brane separation mode, or a compactification deformation. Any source theory that derives the positive-branch determinant must also account for the partner eigenvalue. The same condition governs any electromagnetic-coupling claim. The construction leaves α as a Standard Model input: the scalar normalization, the $U(1)_{\text{em}}$ generator and charge lattice, and the matching pair $(Q_\alpha, \Delta_{\text{th}})$ remain open source-theory data. The electroweak content of the DeVries clue is the pole ratio s_{pole}^2 .

Target III is the route-kernel derivation. Endpoint, interval, and G_2 sources give addresses for a two-channel kernel. In common notation the required reduction is

$$K_J(\lambda) = \begin{pmatrix} \lambda + \Sigma_{hh,J} & \Sigma_{ha,J} \\ \Sigma_{ah,J} & \lambda + \Sigma_{aa,J} \end{pmatrix}.$$

The DeVries entry conditions are

$$\Sigma_{hh,J} = 0, \quad \Sigma_{aa,J} = J, \quad \Sigma_{ha,J} \Sigma_{ah,J} = J.$$

Open-string endpoint data, interval boundary conditions, and localized G_2 gauge sectors provide candidate arenas for such a kernel. The exact DeVries entries remain to be derived from one source theory with a specified field basis and inner product.

The route analysis imposes three necessary consistency conditions. Trace statements stay inside the reduced matrices where they are derived until a shared source operator identifies the trace spaces. The tested Wigner-Eckart/Clebsch parent routes obey their selection rules and miss the parent data required by Eq. (179). An $\text{SO}(32)$ -flavor sector can constrain a completion after its projection, anomaly, hypercharge, global-form, and electroweak-coupling obligations are supplied.

A source route that preserves a common Regge slope can place the DeVries roots as intercepts in

$$M_{N_{\text{osc}},j,\sigma}^2 = \mu^2 x_\sigma(j(j+1)) + \frac{N_{\text{osc}}}{\alpha'} + \Delta_{N_{\text{osc}},j,\sigma}^{\text{Regge}}.$$

The first higher positive sector is $j = \frac{3}{2}$, an algebraic assignment datum. A particle interpretation requires $\mathfrak{A}_{3/2}$: survival projection, gauge representation, chirality or vectorlike status, anomaly constraints, mass-generation rule, production and decay channels, widths, and collider-source check. Appendix C 5 records this filter.

The interval route carries source-derived boundary machinery: variational boundary conditions, modified inner products, vector Robin data, photon zero-mode accounting, and scalar A_5/π_i equations. The G_2 route carries localization and compact-completion machinery, with a scale-placement burden before it reaches pole observables. The electroweak superconnection route packages odd Higgs data and even current data inside one gauge-Higgs complex. The unresolved entry equations in one reduced basis are

$$\Sigma_{ha,J}\Sigma_{ah,J} = J, \quad \widehat{\Sigma}_{aa,J}^{\text{CHM}} = J,$$

with the source scale, reference subtraction, photon projection, and pole map fixed from the same interval data.

The route-kernel comparison combines these two routes. A boundary electroweak superconnection on a CHM interval should give a hatted kernel

$$\widehat{K}_J^{\text{sc/int}} = \Lambda_J^{-2} P_{\text{sc}}^\dagger [K_T^{\text{DtN}} + K_T^{\text{brane}} + \langle \mathcal{F}_{\text{sc}}, \mathcal{F}_{\text{sc}} \rangle_{\text{sc}} - K_\gamma^{\text{ref}}] P_{\text{sc}}.$$

The source-completion conditions are

$$P_W(\Phi_{\text{odd}}) = \frac{3}{4}, \quad P_Z(F_{\text{even}}^{\gamma^\perp}) = 2, \quad P_\gamma(F_{\text{even}}^\gamma) = 0,$$

extra-channel decoupling, the negative-branch scalar functional, and the complex-pole remainder $\Delta_{\text{O1/O10}}$.

The resulting route tests are explicit. Target I gives the pole-placement and high-scale separation test. Target II gives the ordered-assignment source test. Target III gives the common source-kernel entry tests. Target IV gives the scalar or auxiliary interpretation test for the partner branch. Target IX gives the brane-scaling admissibility diagnostic. Target X gives the Hodge/superconnection square-root and breaking-operator test. The W/Z input set is fixed in Sec. III B. Appendix C collects the remaining analytical work as theorem targets.

Target IV is the scalar interpretation of the partner branch. The present paper establishes the algebraic clue, the pole-placement hypothesis, and candidate arenas for the determinant. A dynamical claim for the DeVries quotient requires the two-channel determinant, the ordered electroweak assignment, ray admissibility, the pole-matching theorem, and the controlled scalar interpretation of the negative branch. Global-form and representation checks then constrain any compact completion. Appendix C collects these requirements in theorem-target form.

Appendix A: Algebraic details

We record the elementary algebra. From Eq. (5),

$$x_\pm(J) = \frac{-J \pm \sqrt{J(J+4)}}{2}. \quad (\text{A1})$$

The positive branch has the large- J expansion

$$x_+(J) = 1 - \frac{1}{J} + \frac{2}{J^2} + O(J^{-3}), \quad (\text{A2})$$

while

$$x_-(J) = -J - 1 + \frac{1}{J} + O(J^{-2}). \quad (\text{A3})$$

Thus the positive branch approaches a fixed asymptotic value while the negative branch carries the large negative partner required by the trace.

For $J = 3/4$,

$$x_+\left(\frac{3}{4}\right) = \frac{-3/4 + \sqrt{57}/4}{2} = \frac{\sqrt{57} - 3}{8}. \quad (\text{A4})$$

For $J = 2$,

$$x_+(2) = \frac{-2 + \sqrt{12}}{2} = \sqrt{3} - 1. \quad (\text{A5})$$

Therefore

$$s_{\text{dV}}^2 = 1 - \frac{\sqrt{57} - 3}{8(\sqrt{3} - 1)}. \quad (\text{A6})$$

Appendix B: Numerical reproducibility

The script ‘calculations/devries_spectrum.py’ computes the branch roots and the seed arithmetic value from a supplied M_Z . The central functions are

```
x_plus(J) = (sqrt(J*J + 4*J) - J)/2
sin2_devries = 1 - x_plus(3/4)/x_plus(2)
MW = MZ * sqrt(1 - sin2_devries)
```

The test file ‘calculations/tests/test_devries_spectrum.py’ checks the root identities, the numerical value of s_{dV}^2 , and the seed value obtained from $M_Z = 91.1880$ GeV.

Appendix C: Analytical targets and acceptance tests

The paper uses the DeVries relation as a conditional pole-spectrum proposal. The algebra in Sec. II is exact. The physical claim requires a source theory that supplies the field basis, inner product, normalization, pole map, and decoupling of extra channels. A minimal source package has the form

$$\mathcal{D} : \text{source data} \mapsto (\mathcal{H}_J, K_J(\lambda), \langle \cdot, \cdot \rangle_J, \mathcal{R}_{\text{pole}}), \quad (\text{C1})$$

where $\mathcal{H}_J = \text{span}\{h_{J,a_J}\}$ is the reduced order-parameter/current space. The reduced kernel must satisfy

$$K_J(\lambda) = \begin{pmatrix} \lambda + \Sigma_{hh,J} & \Sigma_{ha,J} \\ \Sigma_{ah,J} & \lambda + \Sigma_{aa,J} \end{pmatrix}. \quad (\text{C2})$$

The entry conditions are

$$\Sigma_{hh,J} = 0, \quad \Sigma_{aa,J} = J, \quad \Sigma_{ha,J}\Sigma_{ah,J} = J. \quad (\text{C3})$$

Equation (C2) is the source-theory version of the two-channel operator $Q(J)$. Every route below is tested against this same kernel target.

1. Target I: pole placement

The pole-placement target is a theorem connecting the reduced source kernel to the complex pole observable

$$s_{\text{pole}}^2 = 1 - \frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2}. \quad (\text{C4})$$

The theorem must specify the convention map from any Breit–Wigner or on-shell input to the complex pole, and it must identify the matching remainder Δ_{match} in

$$s_{\text{pole}}^2 = 1 - \frac{x_+(3/4)}{x_+(2)} + \Delta_{\text{match}}. \quad (\text{C5})$$

The exact low-energy statement requires $\Delta_{\text{match}} = 0$ in the chosen pole scheme. A high-scale, running, or compactification placement is a different source statement with its own running and threshold map.

2. Target II: electroweak assignment and ray admissibility

The ordered electroweak assignment is

$$(J_W, J_Z) = (J_H, J_{\text{adj}}) = \left(\frac{3}{4}, 2\right). \quad (\text{C6})$$

A derivation must produce this order from gauge-Higgs, endpoint, boundary, or compactification data. It must also preserve the electroweak vacuum ray:

$$M_W^2(v) = \frac{g^2 v^2}{4}, \quad M_Z^2(v) = \frac{(g^2 + g'^2)v^2}{4}, \quad M_\gamma^2(v) = 0. \quad (\text{C7})$$

The admissible deformation has one radial parameter v , a fixed photon null direction, and a fixed projective W/Z direction before source-theory matching.

3. Target III: local route kernel

Endpoint, interval, and localized G_2 routes are acceptable only after they produce the entries in Eq. (C2) from a single source problem. The required data are:

1. a two-dimensional light subspace $\mathcal{H}_J$;
2. a physical inner product and normalization scale Λ_J ;
3. a projection that removes photon and extra-channel directions;
4. a pole map from the source spectral parameter to W/Z pole positions;
5. a calculation of $\Sigma_{aa,J}$ and $\Sigma_{ha,J}\Sigma_{ah,J}$.

For an interval route, the source objects may be a Dirichlet-to-Neumann kernel, brane terms, scalar boundary equations, and a Schur complement. For a localized G_2 route, the source objects may be singular-locus gauge data, critical-point wavefunctions, flow-line bilinears, and compact-completion constraints. In either case, trace and determinant statements stay inside the reduced source matrix until a common operator identifies the relevant vector spaces.

4. Target IV: partner branch and scalar data

The negative root belongs to the same diagonalization as $x_+(J)$. A scalar interpretation requires a gauge-invariant functional

$$\mathcal{F}_{\text{sc}}(u_-(J); J) \quad (\text{C8})$$

with a specified normalization, scheme, and source meaning. The functional may be a Higgs-potential parameter, a boundary scalar, a brane-separation mode, a Wilson-line datum, or a compactification deformation. If the source theory assigns $u_-(J)$ to an auxiliary sector, the pole-ratio proposal can survive while scalar claims are removed from the physical conclusion.

The electromagnetic coupling α is a separate endpoint target. A claim for α requires the same source package to fix the $U(1)_{\text{em}}$ generator, charge lattice, scalar normalization, matching scale, and threshold map.

5. Additional filters

Higher branches, Regge placement, and Hodge/superconnection square-root mechanisms are diagnostic filters. A source route that invokes a Regge tower must give a survival projection, gauge representation, anomaly check, mass-generation rule, and pole-width interpretation for every low positive sector. A Hodge or supersymmetric-quantum-mechanics route must produce a normalized two-channel block whose square gives Eq. (C2) after electroweak projection.

The minimum acceptance criterion for the paper is therefore

$$\mathcal{D} \Rightarrow \left[Q(J), (J_W, J_Z) = \left(\frac{3}{4}, 2\right), \right. \quad (\text{C9}) \\ \left. \Delta_{\text{match}} = 0, \mathcal{F}_{\text{sc}} \text{ or auxiliary status} \right].$$

Until Eq. (C9) is derived, the DeVries relation remains a kinematical spectral clue.

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